

**SMART-CONTRACT-BASED TENDER
MANAGEMENT SYSTEM USING BLOCK-CHAIN**

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Dissertation submitted in partial fulfillment of the requirements for the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER ENGINEERING

OF GOA UNIVERSITY



2023-2024

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Bona fide record of work done by

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STUDENT DECLARATION

We hereby declare that the project report titled "Smart Contract based tender management system using blockchain technology" submitted by us to Padre Conceição College of Engineering, Verna in partial fulfillment of the requirement for the award of the degree of Bachelor of Engineering in Computer Engineering under of Goa University is a record of bona fide project work carried out by us under the guidance of Dr. Anusha Pai .

We further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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ABSTRACT

This project introduces an innovative approach to streamline and enhance the tender management process through the integration of electronic wallet technology into a dedicated website. The incorporation of electronic wallet functionality serves as a pivotal feature, promoting seamless financial transactions within the ecosystem. Users can effortlessly manage funds, make re payments, and participate in the tender process with convenience. The integration of an electronic wallet not only simplifies monetary transactions but also ensures a user-friendly and efficient financial ecosystem.

In this advanced technology, the project uses the power of blockchain to revolutionize tender management systems. By implementing a blockchain environment, the project establishes a decentralized and tamper-resistant ledger that guarantees transparency, security, and immutability of data. This technology ensures that every transaction and interaction within the tender management system is securely recorded, creating an permanent trail of activities.

The heart of the project lies in the creation of smart contracts, intelligent self-executing programs that automate and enforce the terms of tender agreements. These smart contracts operate within the secure blockchain environment, ensuring a foolproof and efficient tender process. By automating tasks such as contract execution and payment processing, the system minimizes the risk of errors and fraud, thereby establishing a secured and trustworthy framework. Furthermore, the project encompasses a responsive user interface (UI) design, enhancing accessibility and providing a pleasing and intuitive experience, making the entire tender management process not only secure and intelligent but also user-friendly and visually appealing.

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CHAPTER 1

INTRODUCTION

Current e-tendering processes aren't 'fair and open,' which means that information isn't shared with all parties involved. When a corporation is chosen as the winner of a contract, for example, the information is disseminated 'as they wish. Organisations that bid on a same tender aren't told why their proposal was rejected or why one company was chosen as the winner. A corporation can request this information, but obtaining it is a time-consuming process. Despite the fact that checking these papers is possible, reviewing them takes time. A side from not being transparent, the security of these portals is a major worry, as it can lead to fraud and data manipulation in a central database. If a hacker acquires access to this central database, bids can be shared with competitors, resulting in severe financial and strategic losses for a corporation.

Blockchain technology may be used to address these security problems since it focuses largely on information decentralisation and is safeguarded by encryption combined with an irrefutable block-based architecture for transaction management. As a consequence, Blockchain and Smart Contracts may be used to construct a transparent, decentralised, and secure tendering framework that allows bidders to keep track of portal functionality and all tender portal activity.

Managing payments, funds, and tenders has never been this convenient. The incorporation of electronic wallet functionality serves as a vital feature in promoting seamless financial transactions within the ecosystem. Bidders and Organizations can effortlessly manage funds, make re payments, and participate in the tender process with convenience. The integration of

an electronic wallet not only simplifies monetary transactions but also ensures a user-friendly and efficient financial ecosystem.

1.1 PURPOSE

The project's objective is to design and implement a Smart Contract-based Tender Management System using blockchain technology, with the aim of enhancing transparency, security, and efficiency in the tendering process. This system will be compared to the existing e-tendering systems to assess its effectiveness in reducing fraud, automating contract execution, and ensuring fair competition among bidders. By leveraging blockchain's immutable ledger and smart contract automation, the project seeks to streamline the tendering process, minimize disputes, and provide stakeholders with a tamper-proof and auditable record of all tender-related activities, ultimately promoting trust and cost-effectiveness in the tender process.

1.2 SCOPE OF THE PROJECT

- 1) Designing and developing a robust system that seamlessly integrates blockchain and smart contract technology into the tender management process to enhance transparency and security.
- 2) Ensuring a user friendly interface for stakeholders involved in tender processes to promote user adoption and ease of use.
- 3) The system designed will be using a token based payment system (example: Bitcoin or Ethereum) eliminating any intermediaries like banks or third party, thus preventing frauds and Safeguard the transaction process.
- 4) Integrating a payment wallet into the system which will ensure smooth and seamless transactions between Orgazination and Bidder.

1.3 LITERATURE REVIEW

This literature review explores the dynamic landscape of blockchain-based e-tendering systems through an examination of eight notable projects. This review aims to provide insights into the evolving role of blockchain technology in enhancing transparency, security, and efficiency, it also gives us an idea regarding the challenges that they have faced and the enhancement that we could make.

Title	Author & Publisher	Technologies Implemented / Methodology	Challenges / Advantages	Result / Conclusion
Blockchain Based E-Tendering System	Girija V, Anush R, Charan S, Abhinandan B M, Akhilesh Hiremath Department of CSE Cambridge Institute of Technology, Bangalore, India	1. Tender Officer: Manages tender notifications, downloads and decrypts bidder details from files, and sends confirmation emails. 2. Bidder: Reviews tender notifications, submits details, and checks for responses. 3. Blockchain: Safely stores encrypted data in smaller units using hash codes for added security and for implementing smart contracts.	Challenges: I) Use of tokens: To increase the efficiency and security of payments to successful bidders, the system may implement a token-based payment system. II) Mobile access: To increase convenience and accessibility for stakeholders, the system can be made available on mobile devices Advantages: I) Enhanced Transparency: Its decentralized structure ensures that all participants in the tendering process access the same information, reducing the potential for corruption and fraud. II) Robust Security: Utilizing advanced encryption and blockchain, data becomes highly resistant to alteration or manipulation once recorded. III) Increased Efficiency: Smart contracts streamline procurement procedures, reducing both time and costs.	A blockchain-based electronic tendering system enhances procurement by ensuring transparency, security, and efficiency. It leverages blockchain, smart contracts, and user-friendly interfaces for fair and accountable bidding processes. Future enhancements may include AI, mobile access, and IoT integration, making it a transformative solution that reduces time, cost, and risk in procurement.

Table 1.1 Blockchain Based E-Tendering System.

E-Tendering system using Blockchain Technology	Chandru AS , Shashank Gowda B , Brunda C K , Sumanth L Reddy , Rahul V Dept. of Information Science and Engineering, NIE Institute of Technology, Mysuru, Karnataka, India	1. Blockchain: A decentralized database updated through mining, ensuring data permanence. 2. SHA-256: A highly secure hashing function with near-impossible reversibility and collision resistance. 3. AES Rijndael: A versatile block cipher with choices of key and block sizes. 4. SMTP Protocol: Standard for email transmission and retrieval. 5. NLP: Empowers computers to understand human language through data preprocessing and algorithm development.	<u>Challenges:</u> I) Smart contracts: smart contracts can be used to minimize human interaction and increase efficiency by introducing rules that have to be considered while completing project. II) Use of tokens: To increase the efficiency and security of payments to successful bidders, the system may implement a token-based payment system. <u>Advantages:</u> I) User friendly UI: stakeholders will be able view bidder's profile and also provide on work done. II) Blockchain integrity: Tender details remain immutable, and bids are encrypted, securing the system against manipulation and data exposure.	This paper explores the need for heightened security and transparency in tender management applications. It advocates for the use of advanced encryption techniques like SHA and AES, along with NLP for client feedback analysis. The paper proposes a decentralized blockchain-based system to address security and transparency concerns, detailing the processes and basic implementation steps.
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Table 1.2 E-Tendering system using Blockchain Technology

Smart Tender/Contract Management System Using Blockchain	Thilak K1, Prof. Priya N2 Department of MCA, Jain deemed to be university, Bangalore, India	1. Tender Officer: Tender officers log in, update tender notifications, and can modify or delete them. They download bidder-registered tender files, decrypt data to view bidder information, and send confirmation emails for accepted applications. 2. Bidder: Bidders will login into the account after registration and they will view the tender notifications. If the bidder is okay with tender description he/she will provide their information in text file to the bidder officer. After sending the application they can check the response from the tender officer can make a tender to the tender officer by providing data in a text file. 3. Blockchain: The blockchain is used to store an encrypted format by dividing the data into chunks. Here apply the hash code on chunks for hiding the data after converting it into an encrypted format which is stored in a database.	<u>Challenges:</u> I) Smart Contracts: Smart contracts can be made more secure with more cryptographic algorithms such as SHA-256 to encrypt its confidential contents. <u>Advantages:</u> I) When a company wins a contract, unsuccessful bidders are often left in the dark about the reasons for rejection and the winner's selection. Requesting this information can be cumbersome and time-consuming. II) Centralized tender portals lack transparency and are vulnerable to security breaches, potentially causing fraud. Blockchain, with its decentralized, encrypted, and indisputable transaction management, offers a secure solution.	In applications like tender portals, where paramount transparency and security are vital, traditional technologies fall short. A centralized portal cannot meet the robust security demands. To address these needs, blockchain and smart contracts offer a decentralized, open, and secure solution. This paper outlines the design of such a system, detailing key processes and their fundamental implementations.
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Table 1.3 Smart Tender/Contract Management System Using Blockchain

Blockchain-Based E-Tender Management System	Prof. Vinod Alone, Rushikesh Sose , Tejaswinee Lahare , Pooja Gophane , Afreen Jamadar Vasantdada Patil Pratishthan's College of Engineering, Mumbai, Maharashtra, India	1. Blockchain:Blockchain's decentralization and full replication across nodes make it a distributed database. Transactions trigger mining to update the blockchain, utilizing a consensus protocol and cryptographic hashes to ensure permanence and security. 2. SHA-256: SHA-256, a highly secure hashing function, is mandated by the US government for sensitive data protection. Its security stems from the extreme difficulty of reversing the hash, extremely low collision probability, and the avalanche effect. 3. Natural Language Processing: NLP enables computers to comprehend and process natural language, much like humans. It involves data pre-processing to prepare and clean text data, followed by algorithm development for analysis.	<u>Challenges:</u> I) Use of tokens: To increase the efficiency and security of payments to successful bidders, the system may implement a token based payment system. <u>Advantages:</u> I) Block chain technology is used in the proposed Tender Management System to ensure that the entire tender management process is secure and efficient. Encryption and an irrefutable block-based architecture for transaction management safeguard a block chain. This allows the system to maintain a basic transparent transaction with only the information that the system needs to know.	In conclusion, the Tender Management System revolutionizes Eagle's tender management and contractor interactions with a user-friendly web-based platform. Enhanced security and transparency are achieved through encryption, NLP feedback analysis, and blockchain technology, as discussed in this paper.
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Table 1.4 Blockchain-Based E-Tender Management System

Blockchain-based E-Tendering System	Dhawal Mali, Parth Kitawat, Divya Mogaveera, Mohd. Jawwad Sardar Patel Institute of Technology, Mumbai , India. IEEE	1. Blockchain: Blockchain is a decentralized, distributed database with full replication, where transactions trigger mining and form linked blocks. Attempting to alter a block changes subsequent ones, ensuring immutability. Blockchain's core lies in its implementation and consensus protocol, making it a secure and tamper-proof technology. 2. Smart Contract: A smart contract functions like a blockchain account but is code-driven, not human-controlled. It possesses memory and storage to store data. Being immutable within an immutable blockchain, it ensures unalterable agreement execution between parties. 3. Gas Requirements: It refers to the computational resources needed for blockchain transactions. They align with transaction complexity and determine transaction fees. If gas usage exceeds the gas limit, the transaction fails. In this context, the createTenderFactory process demands the most gas, signifying its intricacy, while rejectBid requires the least. Gas value, measured in GWEI, represents what stakeholders are willing to pay per gas unit, influencing the transaction fee, calculated by multiplying gas requirements and gas value. Complexity and gas needs are directly correlated, as depicted in the graph.	<u>Challenges:</u> The challenges in the given paragraph encompass several key areas of concern. Firstly, maintaining data privacy and encryption are paramount, ensuring sensitive information remains secure. Secondly, bidder verification presents a challenge in efficiently validating credibility and financial stability. Managing bidding periods and deadlines effectively is another challenge to ensure timely submissions. Scalability is crucial to accommodate a growing number of bids and participants. Handling negotiations, bid evaluations, and transparent winner selection criteria all require careful management. <u>Advantages:</u> Firstly, the use of blockchain technology ensures transparency by allowing all bidders access to tender details and evaluation criteria. Secondly, the encryption of documents enhances security, protecting sensitive information. Additionally, the system's asynchronous registration and encryption of bids provide an efficient and secure bidding process. Finally, the release of bidding and negotiation information ensures transparency in winner selection.	In applications like tender portals, traditional technologies fail to meet essential transparency and security needs. Centralized systems can't address the complex security requirements. Blockchain and Smart Contracts offer a solution. This paper discusses the design, processes, and gas requirements of a decentralized tender system. Future research directions include enhancing contract security with advanced cryptographic algorithms like SHA-256 and exploring blockchain's use in other government services.
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Table 1.5 Blockchain-based E-Tendering System

A Design of Blockchain Based Smart Contract for Tendering	Dinies kiyeng , Simon maina Karume , Nelson Masese Department of Computer Science and Information Technology, Kabarak University , Nakuru , Kenya	1. Blockchain is a decentralized and distributed database system characterized by complete data replication, where the creation of new blocks is initiated by transactions, forming a chain of interconnected blocks. Attempting to modify a single block within this chain will result in changes to all subsequent blocks, ensuring the immutability of the entire ledger. The fundamental strength of blockchain lies in its meticulous implementation and the consensus protocol it employs, making it a secure and tamper-resistant technology. 2. Smart contracts, on the other hand, operate as autonomous accounts within the blockchain, driven entirely by code rather than human control. They possess memory and storage capabilities to store data, and they exist in an immutable state within the unchangeable blockchain, guaranteeing the execution of agreements between parties in an irreversible manner. 3. Gas requirements refer to the computational resources necessary for executing transactions on the blockchain. These requirements are directly related to the complexity of the transaction and play a crucial role in determining the associated transaction fees. If the gas consumption surpasses the predefined gas limit, the transaction will fail.	<u>Challenges:</u> Integrating Tokens: In order to enhance the efficiency and security of payments made to successful bidders, the system could incorporate a payment system based on tokens. <u>Advantages:</u> The utilization of blockchain technology in the envisioned Tender Management System serves to guarantee the security and effectiveness of the entire tender management procedure. Through encryption and the indisputable structure of the blockchain for managing transactions, the system establishes a robust security framework. This, in turn, enables the system to maintain transparent transactions, disclosing only the essential information required for the system's operation.	In scenarios such as tender portals, conventional technologies often fall short when it comes to meeting critical transparency and security requirements. Centralized systems struggle to adequately address the intricate security demands of such applications. However, blockchain and Smart Contracts present a viable solution. This paper delves into the design, procedures, and gas needs of a decentralized tender system. Future avenues for research involve strengthening contract security through the adoption of advanced cryptographic algorithms like SHA-256 and investigating the applicability of blockchain in various other government services.
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Table 1.6 A Design of Blockchain Based Smart Contract for Tendering

Contract/ Tendering System Using Blockchain	Ritu Pawar , Rutuja Pohare , Saharsh Wasnik , Sanjay Jengthe , Shubham Paunikar , Shreyash Gondane Department of Information Technology, G H Raisoni College of Engineering , Nagpur, India	1. Tender Officer: Tender officers have the ability to log in to the system, update tender notifications, and have the option to make modifications or deletions to them. They can also download the tender files submitted by registered bidders, decrypt the data to access bidder information, and send confirmation emails to acknowledge accepted applications. 2. Bidder: Bidders are required to log in to their accounts following registration, where they can view the available tender notifications. If a bidder finds a tender description suitable, they can submit their information in a text file to the tender officer. After submitting the application, bidders can track responses from the tender officer and communicate with them by providing data in a text file. 3. Blockchain: The blockchain technology is employed to securely store data in an encrypted format, achieved by breaking down the data into smaller chunks. Hash codes are applied to these chunks to obfuscate the data before it's transformed into an encrypted format, which is then stored in a database.	<u>Challenges:</u> Token Implementation: To enhance the efficiency and security of payments to successful bidders, there is a potential for the system to introduce a payment system based on tokens. <u>Advantages:</u> Blockchain technology plays a pivotal role in the envisioned Tender Management System, ensuring the security and efficiency of the entire tender management process. Through encryption and an indisputable block-based architecture for transaction management, blockchain safeguards the integrity of the system. This results in the system's ability to maintain transparent transactions, disclosing only the essential information necessary for its operation.	In contexts such as tender portals, conventional technologies often fall short in fulfilling the crucial requirements for transparency and security. Centralized systems prove inadequate when it comes to addressing the intricate security demands of such applications. In this regard, blockchain and Smart Contracts emerge as a viable solution. This document explores the design, procedures, and gas requirements of a decentralized tender system. Future research avenues encompass the improvement of contract security through the integration of advanced cryptographic algorithms, such as SHA-256, and the investigation of blockchain's applicability in various other government services.
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Table 1.7 Contract / Tendering System Using Blockchain

Blockchain-based E-tendering Evaluation Framework	Xingbo Gong , Xingyu Taoa , Moumita Dasa , Yuhua Liua , Jack Chenga Department of Civil and Environmental Engineering , Hong Kong University of Science and Technology, Hong Kong SAR	1. Blockchain : Blockchain is used to enhance the e-Tendering process by ensuring transparency, integrity, and security. It creates an immutable ledger for recording tender-related activities, enabling confidential bids, secure evaluations, and transparent outcomes, instilling trust among stakeholders. 2. Smart contracts: The blockchain-based tendering system includes a smart contract, "Record Test," designed in Go language. It offers three main functions: Save, Query, and History Query for efficient bidding process and data storage.	<u>Challenges:</u> The framework only tests some limited scenarios. This design only considers tendering evaluation under standard conditions. Some special requirements, such as quality assessment, have not been validated. <u>Advantages:</u> The system provides increased security compared to traditional centralized databases. The use of blockchain ensures the integrity of bid-related data.	This study introduces a Blockchain-based E-tendering Evaluation Framework to enhance data security in electronic bidding. It employs a transaction data model for information exchange and employs smart contracts for interaction with the blockchain. Although it has shown improved security in certain scenarios, it has limitations, including limited scenario testing and not addressing specialized requirements like quality assessment.
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Table 1.8 Blockchain-based E-tendering Evaluation Framework

Blockchain smart contracts: Applications, challenges, and future trends	Shafaq Naheed Khan, Faiza Loukil, Chirine Ghedira-Guegan, Elhadj Benkhelifa, Anoud Bani-Hani	1. Blockchain: The research paper utilizes blockchain technology, including understanding its principles, architecture, and consensus mechanisms. It likely implements or analyzes smart contracts on various blockchain platforms such as Ethereum or Hyperledger Fabric. 2. Analysis of smart contract solutions: The paper inspects smart contract solutions to observe trends, challenges, and open issues, examining methodologies, technologies, and outcomes for insights.	<u>Challenges:</u> I) Legal Complexity: Smart contracts blur traditional legal boundaries, raising legal questions regarding their enforceability, jurisdiction, and compliance with existing regulations. II) Scalability issue: Ethereum's low transaction throughput leads to congestion and high fees. Research is needed to boost transaction rates. <u>Advantages:</u> I) Automation: Smart contracts automate agreement execution, reducing reliance on intermediaries and simplifies processes. II) Transparency: Transactions recorded on the blockchain are transparent and immutable, enhancing trust and accountability. III) Cost Savings: By eliminating intermediaries and automating processes, smart contracts can reduce transaction costs.	This paper surveys blockchain-enabled smart contracts, emphasizing their transformative potential across industries like finance and healthcare. It categorizes research findings, identifies frequent challenges, and discusses forthcoming trends, serving as a roadmap for future exploration. By addressing technical and usage perspectives, it offers stakeholders invaluable insights into the evolving landscape of smart contract technology, guiding further advancements and promoting a deeper understanding of its implications for decentralized peer-to-peer networks.
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Table 1.9 Blockchain smart contracts: Applications, challenges, and future trends

Blockchain based Smart contracts: A systematic mapping study	Maher Alharby and Aad van Moorsel School of Computing Science, Newcastle University, Newcastle, UK College of Computer Science and Engineering , Taibah University, Medina, KSA	1. Blockchain: records secure transactions shared among participants. Bitcoin introduced cryptocurrencies; Ethereum expanded with smart contracts. Public (Bitcoin, Ethereum) and private (Ripple) blockchains differ in accessibility. 2. Smart contracts: Smart contracts on blockchain execute agreements automatically when conditions are met, promising low fees. They're coded on blockchain, with deterministic contracts not needing external data, unlike non-deterministic ones.	<u>Challenges:</u> I) Security issues: include transaction-ordering, timestamp dependency, mishandled exceptions, re-entrancy, criminal activities, and untrustworthy data feeds, with solutions like code improvements and authenticated data sources. II) Privacy issues: Privacy issues in smart contracts include lack of transactional and data feeds privacy, addressed through tools like 'Hawk' and encrypted requests. III) Performance issue: Sequential execution of smart contracts limits scalability; Vukolić suggests parallel execution for improved blockchain system performance. <u>Advantages:</u> I) Elimination of Third Parties: Smart contracts allow for the execution and enforcement of agreements without the need for a third party, reducing transaction costs and eliminating single points of failure. II) Complex Application Support: Platforms like Ethereum enable the development of complex and customized smart contracts, supporting advanced applications such as decentralized finance (DeFi), digital rights management, and IoT integration.	Blockchain enables trustless communication among parties, facilitating smart contracts. A systematic mapping study identified research gaps: scalability, performance, diverse blockchain platforms, smart contract applications, and criminal activity studies. Most papers focus on identifying and addressing smart contract issues, categorized as codifying, security, privacy, and performance. Future research should address these gaps to advance understanding and development of smart contracts.
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Table 1.10 Blockchain based Smart contracts: A systematic mapping study

Blockchain-Based Smart Contracts in Financial Industry	Pavan S, Dr.Usha J Department of Master of Computer Applications, RV College of Engineering, Bengaluru, India	1. Smart Contracts on Blockchain: The implementation of smart contracts on the Ethereum blockchain to facilitate secure, traceable, and transparent online transactions without the need for intermediaries. 2. Permissionless Blockchain Technology: Utilization of permissionless blockchain technology to manage and record all online transactions, ensuring their consistency and transparency. 3. Systematic Mapping Analysis: A systematic mapping analysis was used to discover and classify peer-reviewed academic papers on smart contracts, identifying research trends and gaps for future work.	<u>Challenges:</u> I) Interoperability and Scalability: Issues with integrating blockchain technology with older systems and scaling the blockchain to handle large volumes of transactions. II) Security Concerns: Ensuring the security of smart contracts against attacks that aim to steal assets or alter contract code remains a significant challenge. <u>Advantages:</u> I) Transaction Security and Efficiency: Blockchain technology enhances transaction security, process identification, process automation, and payment speed, making transactions more efficient and secure. II) Reduced Human Intervention: Smart contracts operate with minimal human intervention, reducing the risk of errors and interference once deployed.	The implementation of blockchain technology and smart contracts in online transactions offers a secure, transparent, and efficient method for managing financial transactions. The study concludes that permissionless blockchain technology can significantly improve transaction security and efficiency, with future enhancements aimed at developing private blockchains for automated claims processing .
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Table 1.11 Blockchain-Based Smart Contracts in Financial Industry

Study of smart contracts	Ms. Shobha tyagi, Gogireddy Chennakesava Reddy, Varikuppala Chandu, Nalabothu Koteswararao Manav Rachna International Institute of Research and Studies, Faridabad, India,	1. Blockchain Technology for Financial Services: Application of blockchain technology, specifically Ethereum, to create a decentralized and secure platform for financial transactions. 2. Smart Contracts for Legal Applications: Exploration of smart contracts within the Ethereum blockchain for automating and enforcing legal agreements. 3. Dynamic Proactive Secret Sharing: Implementation of a proactive secret sharing mechanism to enhance the security of blockchain-based services.	<u>Challenges:</u> I) Regulatory and Technical Issues: Addressing regulatory compliance and overcoming technical challenges such as scalability and security in blockchain implementations . II) Security of Blockchain Networks: Ensuring the security of blockchain networks like Stellar and Ethereum, which are prone to specific vulnerabilities <u>Advantage:</u> I) Decentralization and Transparency: Blockchain provides a decentralized platform that ensures transparency and immutability of transactions, reducing the risk of fraud. II) Efficiency in Legal and Financial Services: Smart contracts automate and streamline processes in legal and financial services, reducing the time and cost associated with traditional methods.	Blockchain technology and smart contracts present a transformative approach to managing financial and legal transactions, offering significant benefits in terms of security, efficiency, and transparency. Future research and development will focus on enhancing blockchain security and scalability to broaden its applications.
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Table 1.12 Study of smart contracts

An Overview of Smart Contract and Use Cases in Blockchain Technology	Mohanta, B.K., Panda, S.S. and Jena, D., 2018, July. An overview of smart contract and use cases in blockchain technology. In 2018 9th international conference on computing, communication and networking technologies (ICCCNT) (pp. 1-4). IEEE.	1. Blockchain Integration: Smart contracts operate on blockchain platforms, enabling decentralized and peer-to-peer transactions without intermediaries. The technology leverages cryptographic hash functions for securing transactions, ensuring they are tamper-resistant and transparent 2. Event-Driven Programming: Smart contracts are programmed to execute specific actions automatically when predefined conditions are met. This event-driven nature ensures that once a smart contract is deployed, it can autonomously enforce agreements and execute transactions. 3. Solidity Programming: The smart contracts are primarily written in Solidity, a high-level programming language designed for developing smart contracts on various blockchain platforms like Ethereum. Solidity allows for defining complex contract behaviors and managing interactions within the blockchain.	<u>Challenges:</u> I) Security Issues: Smart contracts are susceptible to security vulnerabilities, such as bugs in the contract code that can be exploited. Ensuring the security of smart contracts is crucial, as any flaws can lead to significant financial losses. II) Scalability: The scalability of blockchain networks can be a limiting factor for smart contract execution. High transaction volumes can lead to congestion, increasing the time and cost required for contract execution. <u>Advantages:</u> I) Transparency and Trust: Blockchain technology ensures all transactions are recorded on a public ledger, making them transparent and easily verifiable. This transparency enhances trust among participants as all actions taken by the smart contracts are visible and immutable. II) Automation and Efficiency: Smart contracts automate processes that traditionally require manual intervention, reducing the time and cost associated with executing agreements. They streamline operations by executing predefined actions automatically when certain conditions are met, thereby improving overall efficiency	The integration of smart contracts with blockchain technology provides a robust framework for decentralized, secure, and transparent transactions. The event-driven and autonomous nature of smart contracts enhances efficiency and reduces the need for intermediaries, leading to cost savings and faster transaction times. However, challenges such as security vulnerabilities and scalability issues need to be addressed to fully leverage the potential of smart contracts in various applications.
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Table 1.13 An Overview of Smart Contract and Use Cases in Blockchain Technology

Smart Contracts in Blockchain Technology: A Critical Review	Taherdoost , H., 2023. Smart contracts in blockchain technology : A critical review. Information, 14(2), p.117.	1. Smart Contracts on Blockchain: The use of smart contracts within blockchain systems like Ethereum and Hyperledger Fabric to automate, negotiate, and enforce the conditions of agreements without the need for intermediaries. 2. Distributed Ledger Technology (DLT): Utilization of blockchain as a distributed ledger to securely record transactions across a peer-to-peer network, ensuring data integrity and transparency. 3. Cryptographic Techniques: Implementation of cryptographic methods to secure transactions and data within the blockchain, ensuring confidentiality and preventing unauthorized access.	<u>Challenges:</u> I) Immutability and Security: Once smart contracts are deployed on the blockchain, they cannot be altered. This makes it difficult to fix bugs or security issues, leading to potential vulnerabilities. II) Scalability and Data Processing: Mainstream blockchain systems struggle with robust data processing capabilities and efficient execution of smart contracts, leading to limitations in scalability and storage expansion. <u>Advantages:</u> I) Enhanced Trust and Efficiency: Smart contracts reduce the need for intermediaries, thus lowering costs and increasing the efficiency of business processes. They also enhance trust between parties in a decentralized environment. II) Data Integrity: Blockchain technology ensures the integrity of the data stored within it, making it a reliable system for recording transactions and maintaining a distributed ledger.	Smart contracts on blockchain present a promising technology that can revolutionize traditional business processes by enhancing trust and efficiency through decentralized automation. Despite significant challenges related to security, immutability, and scalability, ongoing developments in smart contract languages, data processing capabilities, and security mechanisms are expected to address these issues. Future research and advancements in integrating AI, data science, and game theory into blockchain systems may further enhance the applicability and robustness of smart contracts.
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Table 1.14 Smart Contracts in Blockchain Technology: A Critical Review

A Review of Approaches for Detecting Vulnerabilities in Smart Contracts within Web 3.0 Applications	Li, H., Dang, R., Yao, Y. and Wang, H., 2023. A Review of Approaches for Detecting Vulnerabilities in Smart Contracts within Web 3.0 Applications. <i>Blockchain</i>, 1(1), pp.3-18.	1. Integrated Detection Tools: Tools like EthPloit integrate taint analysis and fuzz testing. By establishing the dependency relationship between variable data and variable control flow in the source code through taint analysis, EthPloit enhances fuzzing test cases, improving path coverage and vulnerability discovery efficiency. 2. Smart Contract Vulnerability Detection System: This system supports the simultaneous detection of specific vulnerabilities in identifier management contracts and common vulnerabilities in general contracts, ensuring reliable security for cyberspace constructed in CoG-MIN. 3. Combination of Symbolic Execution and Machine Learning: This approach reduces contract detection time and improves the accuracy and interpretability of vulnerability detection results. It addresses various challenges in vulnerability detection by leveraging the strengths of both symbolic execution and machine learning methods.	<u>Challenges:</u> I) Security Concerns in Web 3.0: The automated, decentralized, and evolving nature of blockchain technology poses significant security challenges, including accuracy, efficiency, and adaptability to emerging vulnerabilities. II) Complexity of Managing Multiple Identifiers: Managing various identifiers (identity, content, IP, and domain names) in a flexible and unified manner presents significant challenges, particularly in ensuring security and efficiency before the chaining review. <u>Advantages:</u> I) Enhanced Detection Performance: Integrated detection tools improve vulnerability coverage and detection accuracy by combining the strengths of different methods and addressing their individual weaknesses. II) Improved Security for Cyberspace: The smart contract vulnerability detection system provides reliable and efficient security for identity management contracts, supporting the secure construction of a co-governed network space community.	The integration of advanced detection techniques, such as symbolic execution and machine learning, along with tools like EthPloit, significantly enhances the efficiency and accuracy of vulnerability detection in smart contracts. These integrated approaches address the limitations of individual methods, offering a comprehensive solution to the security challenges posed by the decentralized and evolving nature of blockchain technology in the Web 3.0 era. This ensures robust and reliable security for various applications within the cyberspace, supporting the development of a co-governed network community.
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Table 1.15 A Review of Approaches for Detecting Vulnerabilities in Smart Contracts within Web 3.0 Applications

Experiences Using Ethereum and Quorum Blockchain Smart Contracts in Dairy Production	Melissari, F., Papadakis, A., Chatzitheodorou, D., Tran, D., Schouteten, J., Athanasiou, G. and Zahariadis, T., 2024. Experiences Using Ethereum and Quorum Blockchain Smart Contracts in Dairy Production. Journal of Sensor and Actuator Networks, 13(1), p.6.	1. Blockchain-based Traceability System: The study developed a traceability system using Ethereum and Quorum blockchains. This system models the end-to-end production process of PDO feta cheese, ensuring the authenticity and compliance with legislation by implementing smart contracts. 2. Smart Contracts: Legislative rules were codified into smart contracts to automatically enforce compliance and record data immutably. These contracts ensure food authenticity and provide operational parameters such as pH, temperature, and maturation duration. 3. Dual-Chain System: A dual-chain approach was used to handle large and dynamic data volumes. Transactions were performed on a private Quorum blockchain, which was anchored to the public Ethereum blockchain to optimize resource use, enhance transaction speed, and minimize costs.	<u>Challenges:</u> I) Data Volume and Transaction Speed: Handling large and dynamically changing volumes of data efficiently while maintaining the speed and cost-effectiveness of blockchain transactions was a significant challenge. II) Consumer Trust and Verification: Ensuring that consumers could access and verify traceability information effectively required a robust and user-friendly interface, and the integration of complex production parameters into the blockchain system. <u>Advantages:</u> I) Immutability and Transparency: The blockchain technology provided a permanent and tamper-resistant record of data, enhancing transparency and trust among consumers and stakeholders. II) Decentralization and Efficiency: The decentralized nature of blockchain reduced the risk of single points of failure and allowed for more efficient data management and traceability across the supply chain.	The study successfully developed and implemented a blockchain-based traceability system for PDO feta cheese production, which enhanced food safety, authenticity, and consumer trust. The dual-chain approach and the use of smart contracts ensured efficient data handling, compliance with regulations, and transparent communication of traceability information to consumers. This system represents a significant advancement in combating food fraud and improving traceability in the dairy industry.
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Table 1.16 Experiences Using Ethereum and Quorum Blockchain Smart Contracts in Dairy Production

As we go deeper into the world of blockchain-based e tendering systems, it is clear that each project takes its own approach. The literature review covers a variety of studies on blockchain-based smart contracts, emphasizing their applications, advantages, and challenges. One study highlights that smart contracts automate processes, increase transparency, and reduce costs, but face legal complexities and scalability issues, particularly in finance and healthcare applications [Table 1.1 to Table 1.8]. Another research focuses on the elimination of intermediaries and support for complex applications, yet identifies significant challenges such as security issues, privacy concerns, and performance limitations [Table 1.9 and Table 1.10]. In the financial sector, smart contracts bring transparency and efficiency but are hampered by security vulnerabilities and scalability problems [Table 1.11 to Table 1.13]. A critical review highlights the enhanced trust, efficiency, and data integrity provided by smart contracts, while also pointing out the difficulties related to immutability, security, and scalability [Table 1.14 to Table 1.16] .

Despite the competition and its successes, it is important to emphasize the uniqueness of our project and how it differs from others. Unlike traditional methods, our project goes one step further and prioritizes user experience. We recognize that user engagement and adoption is crucial to the success of any system, so we place great emphasis on creating user experience that is not only simple but also attractive. We aim to get on board all the stakeholders by ensuring participants find our platform easy to navigate and engage with. While other projects use normal payment methods to increase efficiency, our project goes one step further by integrating an e-wallet into the website itself, so as to make the user experience much better. This approach not only ensures the integrity and confidentiality of information, but also increases stakeholder trust, thus reducing the risk of management or information leakage. While we are aware of scalability issues, our system is designed to accommodate a number of tenders and participants. This strategic approach ensures that our platform remains

functional and adapts to changing needs. De-centralized approach using blockchain also ensures that the stakeholders trust the website and start , advancing towards this approach for tender creation and the bidding process. We believe that our project, which emphasizes user experience, uses the latest technology and is able to do this, provides a solution problem that sets new standards for Smart contract tender management using blockchain.

1.4 BENCHMARK & HURDLES

BENCHMARK:

After going through the above research papers, the following benchmarks have been set:

- 1) Develop Payment Wallet:** Each bidder will have his electronic wallet linked to his account on the website, once the tender has been assigned to a particular bidder, a smart contract will execute ensuring smooth and secure transaction.
- 2) Provide Global Accessibility through blockchain :** Blockchain networks operate globally, allowing for seamless cross-border transactions and collaborations without the need for currency conversion or traditional banking intermediaries.
- 3) Ensure Immutable Record-Keeping:** Tender office personnel as well as the bidders will not be able to manipulate any data related to the proposal once it is issued till the end of the duration of the bidding period.

HURDLES:

During the development of the project, the following issues may arise:

- 1) Scalability Concerns:** The use of public blockchains may lead to scalability issues, particularly in cases with a high volume of transactions, potentially causing delays.
- 2) Inflexibility:** Smart contracts, once deployed, are difficult to modify, potentially limiting adaptability to changing tender requirements.

3) Adoption Challenges: Convincing all stakeholders to embrace this technology may be a long-term challenge, especially in sectors where conservative procurement approaches prevail.

1.5 CHAPTER SUMMARY

- 1) The introduction outlines the development of a blockchain-based Tender Management System for transparency and security in the tendering process.
- 2) The project scope includes integrating blockchain, smart contracts, a user-friendly interface, and a token-based payment system to enhance efficiency.
- 3) Literature survey identifies blockchain benefits in e-tendering, highlighting challenges such as token implementation and technology adoption.
- 4) Benchmarks involve creating a payment wallet, ensuring global accessibility through blockchain, and maintaining an immutable record-keeping system.
- 5) Hurdles in the project include potential scalability issues, smart contract inflexibility, and convincing stakeholders to adopt blockchain technology.
- 6) The purpose is to streamline the tendering process, reduce fraud, and provide a tamper-proof record through blockchain's transparency and smart contract automation.

CHAPTER 2

OVERALL DESCRIPTION

The "Smart-Contract-Based Tender Management System Using Blockchain" project is a revolutionary initiative designed to transform the traditional tendering process by harnessing the power and security of blockchain technology. This system employs smart contracts, which are self-executing contracts with the terms of agreement directly written into lines of code. By utilizing blockchain's immutable and transparent nature, the system ensures a tamper-proof record of all tender-related activities, bids, and contracts. Bidders and Organizations involved in the tendering process can experience enhanced trust, reduced administrative overheads, and minimized risks of fraud or manipulation.

Furthermore, the integration of smart contracts automates various stages of the tender process, streamlining operations, and ensuring that all participants adhere strictly to predefined rules and timelines. Overall, the project aims to bring efficiency, transparency, and security to the tender management landscape, setting a new standard for public and private sector tender.

2.1 PROJECT PERSPECTIVE

Our project offers a transformative perspective on traditional tendering processes by leveraging the inherent benefits of blockchain technology. At its core, the project aims to enhance transparency, security, and efficiency in tender management. By employing smart contracts, which are self-executing contracts with the terms of agreement directly written into code, the system ensures that all parties involved adhere to predefined conditions without the need for intermediaries. This not only minimizes the potential for disputes but also significantly reduces administrative overheads. Furthermore, the immutable nature of blockchain ensures that all transactions and bids are recorded transparently and cannot be

tampered with, thereby promoting trust among stakeholders. Ultimately, this innovative approach has the potential to revolutionize tendering processes, making them more streamlined, trustworthy, and resilient to fraud or manipulation.

2.2 PROJECT FUNCTIONS

- 1) Profile Creation / Setup:** Enter organization / company details, assign a technical account handler for oversight.
- 2) Tender Viewing:** The bidder can monitor completion progress and address malpractices leading to potential tender cancellation.
- 3) Bid Submission:** Browse available tenders, submit bids with detailed estimates and document attachments.
- 4) Approval Status:** Post-deadline notifications with reasons for bid acceptance or rejection.
- 5) Work Progress Tracking:** Monitor ongoing projects, update task statuses, and communicate with the organization.
- 6) Payment Management and Processing:** Receive notifications for payment milestones, initiate payment releases and track transaction histories.
- 7) Tender Creation:** Draft tender descriptions, set bidding and project timelines, and attach necessary documentation.
- 8) Tender Management:** Decide on tender continuation based on bid evaluations, handle bidder communications, and finalize contracts.
- 9) Tender Evaluation:** Assess project progress, ensure quality and standards are maintained.
- 10) Feedback mechanism:** Provide feedback for the work done.

iv) Approval Check: Once the deadline for tender has been passed, the bidder will receive a notification regarding whether he has been selected or rejected. If the bid is rejected, then he will be provided with an appropriate reason by the Tender Organization. If the bid is selected, then he will be directly proceeded towards the smart-contract.

2) Organization

i) Create User Profile: On signing up to the website, the organization will have to create his profile in which he will have to specify organization registration number, his workforce, the name of the organization, the organization address, electronic wallet address, public address. He will have to assign 1 of its organization worker who has technical knowledge as the account handler who will have the access control over the account.

ii) Create Tender: For creating the tender, the organization should give a small description about the tender, the tender motive, the tender locale, duration of the bidding period as well as duration of the tender period, type of tender (Government or Private), the organization can also attach pdf containing all the details and some additional details as well such as materials required.

iii) Manage Tender: The organization has the authority to either cancel the tender before the bidding period ends. The organization can also cancel the tender after the bidding period ends if it doesn't find any satisfactory bids for the tender. If it finds satisfactory bids, then the organization can assign the tender contract to that particular bidder and can proceed to the smart contract.

iv) Evaluate Tender: After assigning works to bidders, the organization can check how much of the work has been completed, the organization can also terminate the ongoing work if it finds any malpractices. It can also provide feedback to the completed work.

2.4 OPERATING ENVIRONMENT

- 1) **Blockchain Platform:** By using blockchain, we ensure that tender records are tamper-proof, reducing the risk of fraud and enhancing trust among project stakeholders. Ethereum is commonly used for its robust smart contract capabilities.
- 2) **Smart Contracts:** Will be developing smart contracts using programming languages which is Solidity (for Ethereum) because it is compatible with EVM (Ethereum-Virtual-Machine).
- 3) **Cryptocurrency Integration:** Implement cryptocurrency support, for payments within the system. This involves integrating cryptocurrency wallets and payment gateways.
- 4) **Web Development:** Build a user-friendly web interface for tender creation, submission, evaluation, and contract execution. Utilize technologies such as HTML, CSS, JavaScript, and popular web frameworks like React.JS and Node.JS.
- 5) **Database:** Implement a database system (MongoDB) to store tender-related data, user information, and transaction history. MongoDB is used to store unstructured data in JSON format.
- 6) **Security protocols:** Implementing various algorithms to protect against vulnerabilities and ensure data privacy and integrity.
- 7) **Testing Tools:** Sepolia Testnet using RemixIDE is used as testing frameworks will be used for testing to ensure the correctness of smart contracts.

2.5 CHAPTER SUMMARY

- 1) The project uses blockchain and smart contracts to make tender processes more trustworthy.
- 2) It aims to simplify tendering, reduce fraud, and lessen administrative work.
- 3) Users can create profiles, view tenders, submit bids, and track approvals.

- 4) Two main user roles are "Bidder" (bid submission, tender viewing) and "Organization" (tender creation, management).
- 5) The system focuses on transparency, security, and efficiency, with a feedback feature for better communication.
- 6) Operating on Ethereum, using Solidity for smart contracts, and MongoDB for data storage ensures a strong foundation.

CHAPTER 3

REQUIREMENTS

In this chapter we will learn about different types of Functional and Non-Functional requirements as well as Hardware and Software requirements which ensure compatibility, aiding smooth project execution by setting clear expectations and guidelines for development and performance.

3.1 FUNCTIONAL REQUIREMENTS

- 1) **User Registration:** Users should be able to create accounts and securely link their electronic wallets for payment purposes.
- 2) **Tender Creation:** Allow authorized users (e.g., organizations or government agencies) to create and publish tenders with detailed specifications, requirements, and deadlines.
- 3) **Tender Submission:** Bidders or contractors can submit their bids and proposals electronically through the platform, ensuring transparency and eliminating physical paperwork.
- 4) **Smart Contract Automation:** Smart contracts should automatically execute key tender-related actions, such as contract formation, and payment release based on predefined rules and conditions.
- 5) **Payment in Cryptocurrency:** Enable secure and efficient payments using cryptocurrencies like Bitcoin, Ethereum, or other supported tokens through integrated wallets and payment gateways.
- 6) **Blockchain Encryption:** Blockchain encryption is a security feature that uses cryptographic techniques to protect data within a blockchain, ensuring its integrity and confidentiality.

- 7) **Real-time Notifications:** Keep users informed about tender updates, bid status changes, and important deadlines through in-app notifications.
- 8) **Document Management:** Allow for the secure uploading, storage, and sharing of tender-related documents, ensuring easy access for all involved parties.
- 9) **Contract Management:** Facilitate the creation, execution, and management of contracts through smart contracts, automating payment milestones and contract amendments.
- 10) **Mobile Accessibility:** Develop mobile applications or ensure responsive web design for easy access on smartphones and tablets.

3.2 NON-FUNCTIONAL REQUIREMENTS

1. Security:

- i) **Immutability:** Ensure that once a tender or contract is recorded on the blockchain, it cannot be altered or tampered with.
- ii) **Data Encryption:** All sensitive data, including bidder details and tender specifications, must be encrypted both at rest and in transit.
- iii) **Access Control:** Implement strict access controls to ensure only authorized individuals can view, modify, or execute contracts.

2. Performance:

- i) **Transaction Speed:** The system should process and validate transactions quickly to maintain an efficient tendering process.

3. Usability:

- i) **User Interface:** The system should have an intuitive and user-friendly interface for bidders and organizations.

4. Interoperability:

i) Blockchain Compatibility: Ensure that the chosen blockchain platform and smart contract languages are compatible and future-proof.

5. Auditability and Transparency:

i) Transaction Traceability: Every transaction and modification on the blockchain should be traceable to maintain transparency.

ii) Audit Logs: Maintain comprehensive logs of all activities, ensuring that every action can be audited if required.

3.3 HARDWARE REQUIREMENTS

1) Windows 7 or higher = Using Windows 7 or newer versions ensures broad software compatibility, allowing developers to leverage a diverse range of tools and platforms. While Windows 7 lacks ongoing support, newer versions like Windows 8.1 and 10 provide enhanced security features, including built-in antivirus protection and regular updates, which are vital for safeguarding sensitive blockchain data and transactions.

2) I3 processor system or higher = Using an Intel i3 or higher processor ensures faster transaction verifications and efficient cryptographic computations. This enhanced processing power streamlines blockchain operations and offers developers a responsive environment for optimal performance and testing.

3) 4 GB RAM or higher = Utilizing systems with 4 GB RAM or higher RAM ensures enhanced performance and efficiency. With ample memory, these projects can manage larger datasets and execute complex tasks with greater speed, streamlining both development processes and deployment phases, thereby encouraging smoother operations and optimal outcomes.

4) 100 GB ROM or higher = Employing systems with a 100 GB ROM or higher ROM are advantageous due to the ample storage capacity they provide. Blockchain applications, often

dealing with substantial transaction data and smart contracts, benefit from the increased space, ensuring efficient operations, swift data access, and scalability as the project expands.

3.4 SOFTWARE REQUIREMENTS

1) Node JS = Using Node.js for blockchain projects is advantageous due to its event-driven architecture, efficient handling of asynchronous operations, and extensive package ecosystem from npm. It allows for rapid development, scalability, and seamless integration with other frameworks and databases.

2) React JS = Using ReactJS for blockchain projects streamlines the development of dynamic and user-friendly interfaces. Its component-based approach enhances reusability and simplifies complex UI tasks, while seamless backend integration ensures efficient interaction with blockchain networks, making ReactJS an ideal choice for building scalable and responsive blockchain applications.

3) Solidity = Using Solidity for smart contract projects is beneficial due to its specialized design for the Ethereum blockchain. Solidity offers a familiar and robust framework for developers, ensuring efficient and secure contract deployment. Its high-level language facilitates the creation of complex decentralized applications while minimizing potential vulnerabilities, enhancing trust in the blockchain ecosystem.

4) Remix IDE = Using Remix IDE for smart contract projects offers a streamlined development experience, providing a comprehensive environment with built-in tools for writing, testing, and deploying smart contracts. Its integrated debugger and static analysis tools enhance code reliability and security, making it easier for developers to identify and fix potential issues efficiently.

5) PostgreSQL = PostgreSQL supports secure data storage, integrity, and advanced querying, making it indispensable for managing tender processes within blockchain-based smart contracts.

6) Visual Studio Code = Using Visual Studio Code (VS Code) is beneficial for frontend development due to its lightweight yet powerful interface, extensive library of extensions, and robust debugging capabilities, enabling developers to create efficient and visually appealing user interfaces with ease.

7) MetaMask = MetaMask streamlines electronic wallet transactions, offering a secure bridge to decentralized apps. Its intuitive interface enhances user experience, facilitating easy navigation within the decentralized finance landscape.

8) Sepolia TestNet = Sepolia testnet is used to deploy, test, and verify smart contracts in a safe environment. This allows developers to identify and fix issues without incurring costs or risks on the testnet as well as mainnet, ensuring a secure and reliable tender management system.

3.5 CHAPTER SUMMARY

1) The project covers user registration, tender creation, smart contract automation, cryptocurrency payments, blockchain encryption, real-time notifications, document, and contract management.

2) Functional aspects include secure user registration, tender processes, smart contract automation, cryptocurrency payments, real-time notifications, and mobile accessibility.

3) Non-functional requirements involve security, performance, usability, interoperability, and auditability.

4) Hardware specifications include Windows 7 or higher, i3 processor system or higher, 4 GB RAM or higher, and 100 GB ROM or higher.

5) Software components encompass Node.js, ReactJS, Solidity, Remix IDE, PostgreSQL, Visual Studio Code, MetaMask, and Sepolia TestNet.

CHAPTER 4

SYSTEM DESIGN

4.1 SYSTEM DESIGN DIAGRAM

The implementation of system design is a significant percentage of the overall design cycle. It is critical that the implementation of the design be handled as efficiently as possible. The decisions before and during the design implementation phase can have a dramatic impact on the implemented design and project schedule. The most important design implementation tasks are presented in this chapter. Figure 4.1 presents a system design implementation flow.

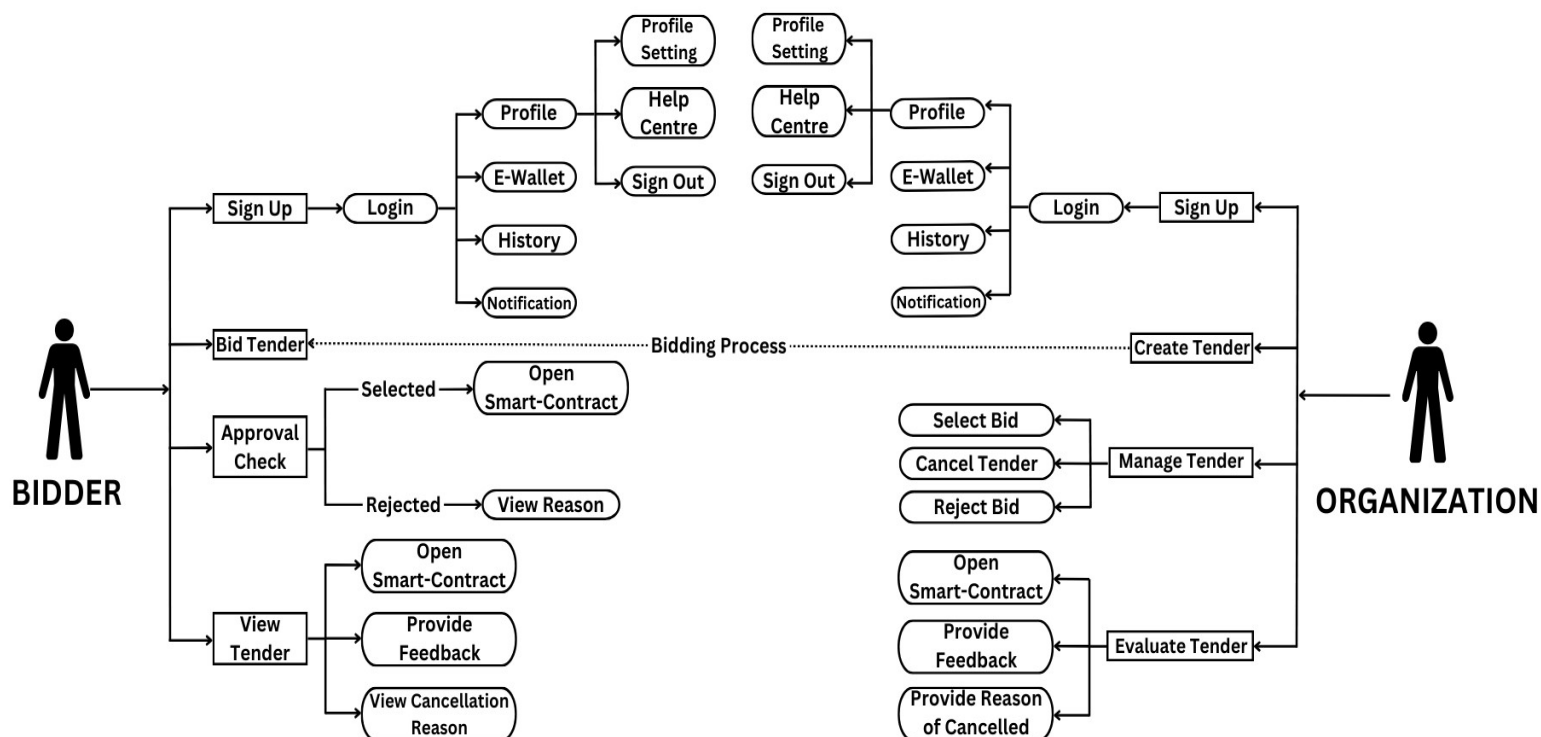


Fig. 4.1 System Design

The above Fig 4.1 displays all the functionalities of all the stakeholder namely Bidders and Organizations. It gives us a brief introduction of the entire system and its working.

I) Actors:

1) Bidder:

- Represents an individual or company participating in the tender process.
- Interacts with the system to bid for tenders, check for approvals, and view tender.

2) Organization:

- Represents the company that initiates and manages the tender process.
- Interacts with the system to publish tenders, manage tenders, and evaluate tender.

II) Use Cases:

1) BidTenders: Bidder can view a list of available tenders and place their bid for any tender.

2) Check for approvals: Bidder can submit a bid for a specific tender and check if they have selected or rejected.

3) View Tender: Involves interaction with smart-contracts, providing feedback and also viewing the tender cancellation reason.

4) Create Tender: Organization can create and publish a new tender for bidding.

5) Manage Tender: Provides the Organization with options such as select bid, reject bid or cancel tender.

6) Evaluate Tender: Involves interaction with smart-contracts, providing feedback and also tender cancellation.

III) Smart Contracts: Smart contracts are programmable contracts that enforce the terms of agreements in the blockchain. They are implicitly represented in the use case diagram.

IV) Notifications: Interaction beyond the system boundaries is shown with notifications. Highlights the need for communication between the organization and the successful bidder.

V) Profile: It gives a brief introduction about the Bidder and Organization details.

VI) Electronic Wallet: It is used as an enhanced payment method for secure and trustworthy transaction between the 2 entities.

VII) History: It displays all the tenders that the Organization has published or the Bidder has bidden for.

4.2 CHAPTER SUMMARY

In summary, a use case diagram provides a high-level overview of the interactions between actors which are Bidders and Organizations and the system's functionalities in a smart contract-based tender management scenario. It serves as a valuable tool for communication and understanding the key features and interactions within the system.

CHAPTER 5

IMPLEMENTATION AND DESIGN

This chapter discusses the system implementation and design phase: making the new system available to a prepared set of users (the deployment), and positioning on-going support and maintenance of the system within the transition. At a finer level of detail, deploying the system consists of executing all steps necessary to place the newly developed system into convention, confirming that all data required is available and accurate, and validating that recruitment functions interact with the system are functioning properly.

5.1 UI DESIGN

This chapter is all about UI Design. We're talking about making digital things look good and easy to use. It's like putting colors and shapes together to make stuff on screens awesome and user-friendly. So, let's get into it and learn how to make things look cool and simple!



Fig. 5.1 Home Page Design.

A landing page for the website shown in Fig. 5.1 is basically is built to encourage users to perform actions such as signing up and even subscribing to the platform.

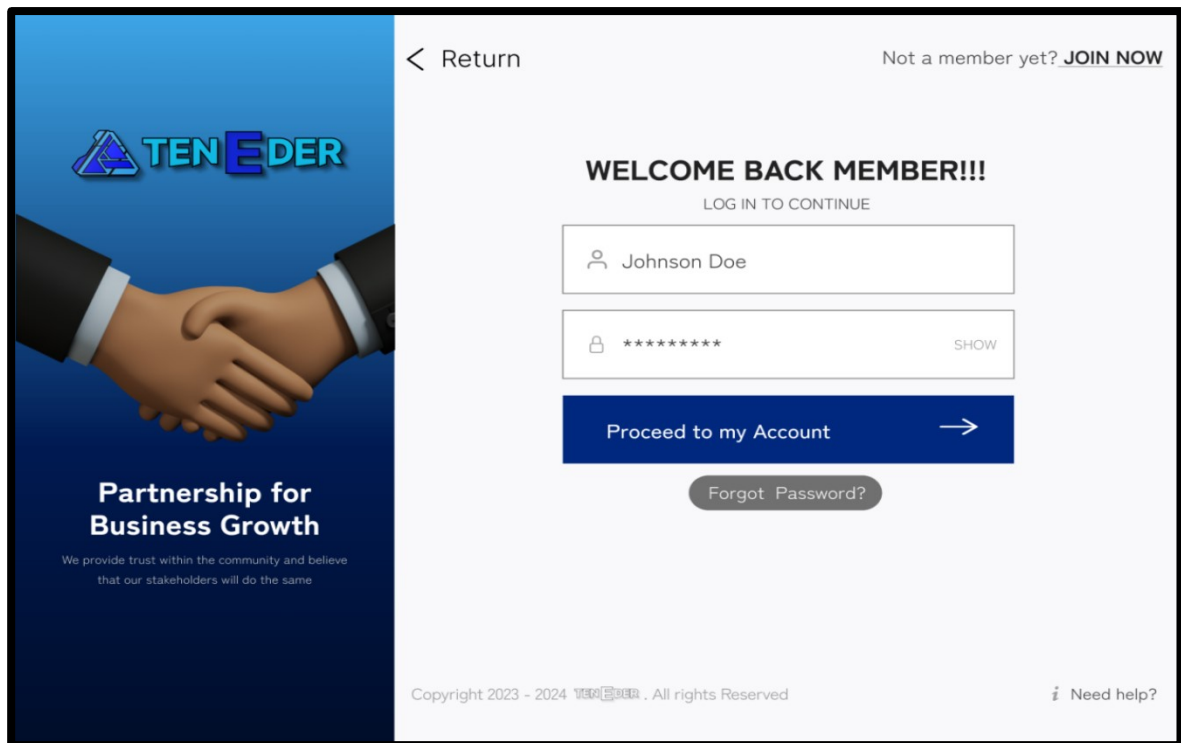


Fig. 5.2 Login for Bidder.

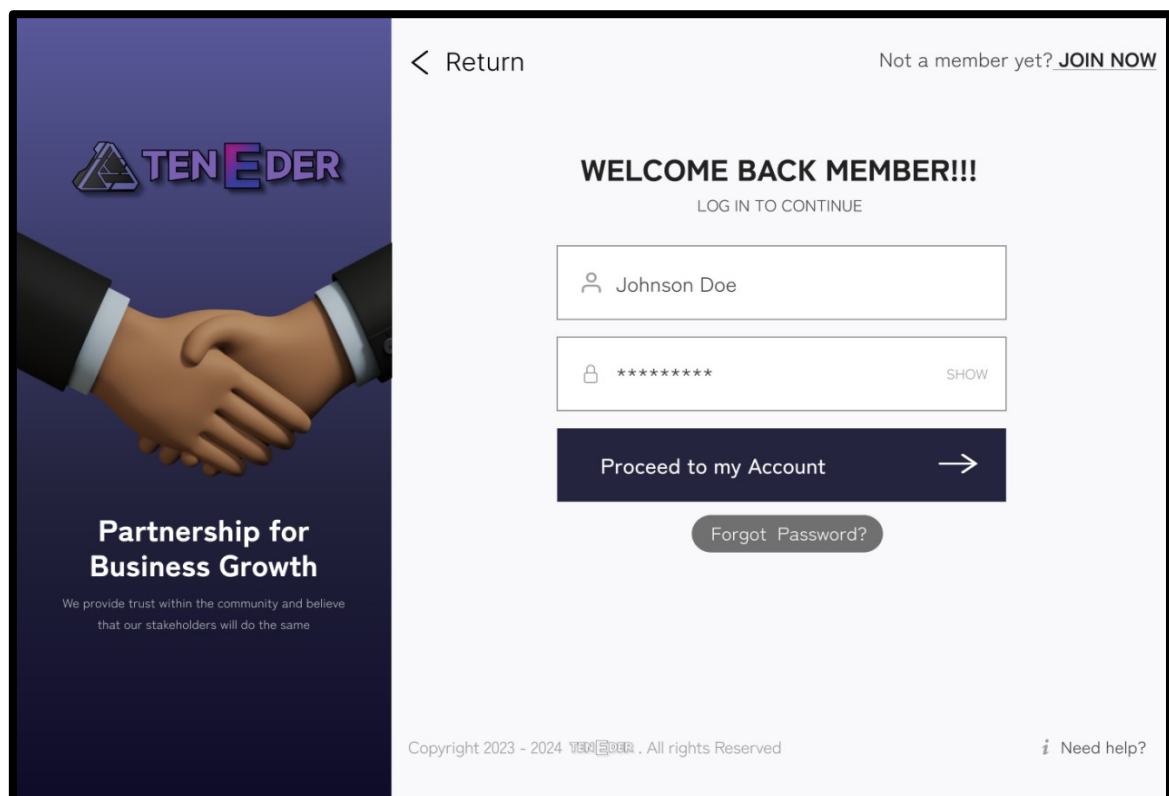


Fig. 5.3 Login for Organization.

Fig. 5.2 and Fig. 5.3 are login pages which contain a basic form of authentication and is used by various stakeholders to log into their account.

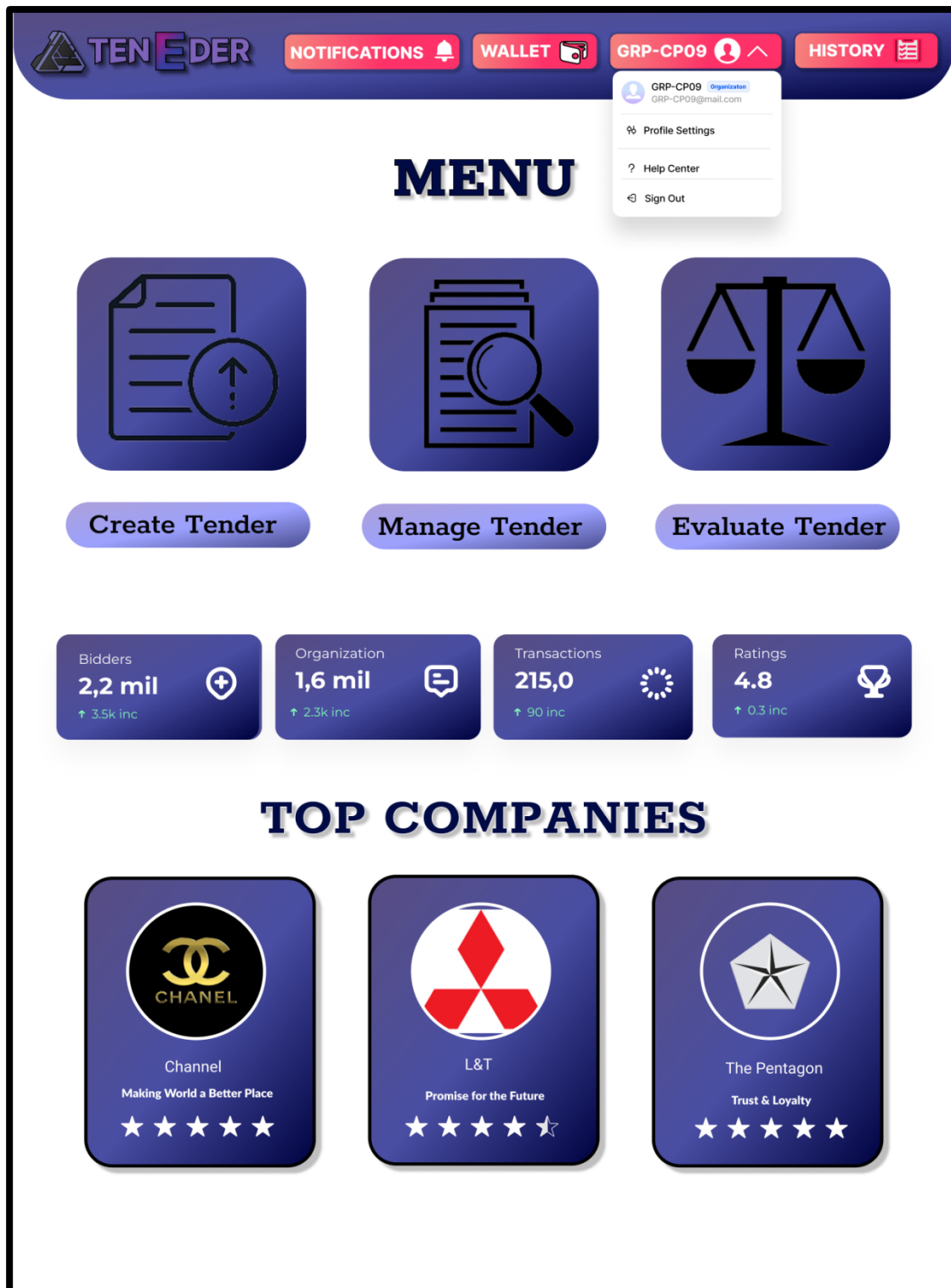


Fig. 5.4 Dashboard for Organization.

The Fig. 5.4 displays a dashboard containing the various functions that can be performed by the organization for tender activities. These include tender creation, managing tender and evaluating tender.


NOTIFICATIONS
WALLET
GRP-CP09
HISTORY

< Return

CREATE TENDER

*** DESCRIPTION ABOUT THE TENDER**

Ex : Facility Management And Maintenance At The Building Of The Ministry Of Communications
And Multimedia pernem. Plot no. 234/2a/4.

*** TENDER MOTIVE**

Ex : Bridge tender aims to unite cities, boost commerce, enhance transportation, and foster regional growth through connectivity.

*** TENDER LOCALE**

*** Address Line 1**

Address Line 2

Address Line 3

*** City**

*** State**
-Please select-

*** Postal Code**

*** DURATION OF BIDDING PERIOD**

*** DURATION OF WORK PERIOD**

*** TYPE OF TENDER**
-Please select-

*** APPENDICES**
Attach PDF
Ex : YTL.pdf

MATERIALS REQUIRED

⇒

⇒

⇒

⇒

⇒

⇒

PUBLISH THE TENDER

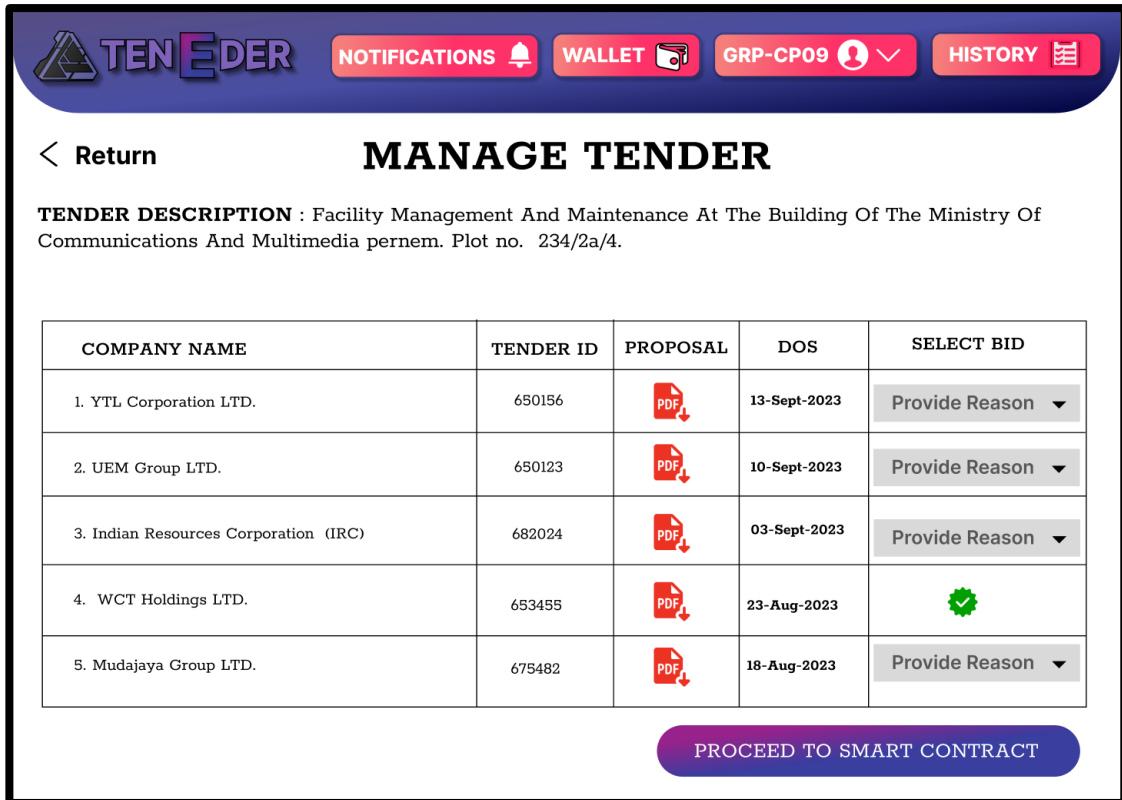
Fig. 5.5 Create Tender Page.

The tender creation form shown in Fig. 5.6 is a function that an organization can perform to create various tenders by specifying the requirements such as tender location, bidding period, work period etc.

 NOTIFICATIONS  WALLET  GRP-CP09  HISTORY 			
< Return MANAGE TENDER			
TENDER DESCRIPTION	TYPE	DUE DATE	CANCEL
1. Contractor. Procurement and construction of multistoried residential finished houses for economically weaker sections (ews). No. 4567832  Panjim  India	Government	22 Days to go 15-Feb-2024	
2. Management Services Of An Apartment House , opp district court. No. 7899020  Mapusa  India	Government	42 Days to go 07-Mar-2024	
3. Facility Management And Maintenance At The Building Of The Ministry Of Communications And Multimedia pernem. Plot no. 234/2a/4. NO. 48802666  Pernem  India	Government	Concluded 	
4. Electrical System Installation For Unimap-construction Of University Of Vasco ,opp Health Center Building (tender No . Pers/r-pk (4) 1511/15-2022) No. 48489530  Vasco  India	Government	88 Days to go 26-Apr-2024	

Fig. 5.6 (i) Manage Tender for Organization.

The organization can view all the details about the tenders that they have uploaded and for each tender a unique tender id is created as shown in Fig. 5.6 (i).



TENDER DESCRIPTION : Facility Management And Maintenance At The Building Of The Ministry Of Communications And Multimedia pernem. Plot no. 234/2a/4.

COMPANY NAME	TENDER ID	PROPOSAL	DOS	SELECT BID
1. YTL Corporation LTD.	650156		13-Sept-2023	Provide Reason ▼
2. UEM Group LTD.	650123		10-Sept-2023	Provide Reason ▼
3. Indian Resources Corporation (IRC)	682024		03-Sept-2023	Provide Reason ▼
4. WCT Holdings LTD.	653455		23-Aug-2023	
5. Mudajaya Group LTD.	675482		18-Aug-2023	Provide Reason ▼

PROCEED TO SMART CONTRACT

Fig. 5.6 (ii) Manage Tender for Organization.

An organization will be able to view all the bids for a particular tender only after the bidding period is over as shown in Fig. 5.6 (ii) and can select anyone bid for further process of smart contract.

TENEDER NOTIFICATIONS WALLET GRP-CP09 HISTORY

MANAGE TENDER

TENDER DESCRIPTION : Facility Management And Maintenance At The Building Of The Ministry Of Communications And Multimedia pernem. Plot no. 234/2a/4.

NO. 48802646 TYPE : Government DOC : 23-Sept-2023

COMPANY NAME	DOS	SELECT BID
1. YTL Corporation LTD.	13-Sept-2023	✓
2. UEM Group LTD.	10-Sept-2023	✓
3. Indian Resources Corpor	03-Sept-2023	✓
4. WCT Holdings LTD.	23-Aug-2023	✓
5. Mudejaya Group LTD.	18-Aug-2023	✓

Are you sure you want to cancel this tender ?

☹️

Tender data

Company name:

Tender ID:

Cancellation Reason:

CANCEL **RETURN**

CANCEL THE TENDER

Fig. 5.6 (iii) Manage Tender for Organization.

An organization can also cancel the tender by providing appropriate reason like work not required or if they didn't find any satisfactory bids as shown in Fig 5.6 (iii) .

TENEDER NOTIFICATIONS WALLET GRP-CP09 HISTORY

REASON FOR CANCELLATION OF TENDER

Tender

Name of the Organization: Panjim Municipal Council
GST Number: 035468276265
Tender ID: 4567832
Name of the Bidder: L & T
Amount: 0.90 ETH
Duration: 26/10/2023 - 29/11/2023
Status: **Cancelled**

SUBMIT

Fig. 5.7 Tender Cancellation for Organization.

The Fig. 5.7 displays the tender cancellation process done by the organization if it finds any discrepancies related to work done.

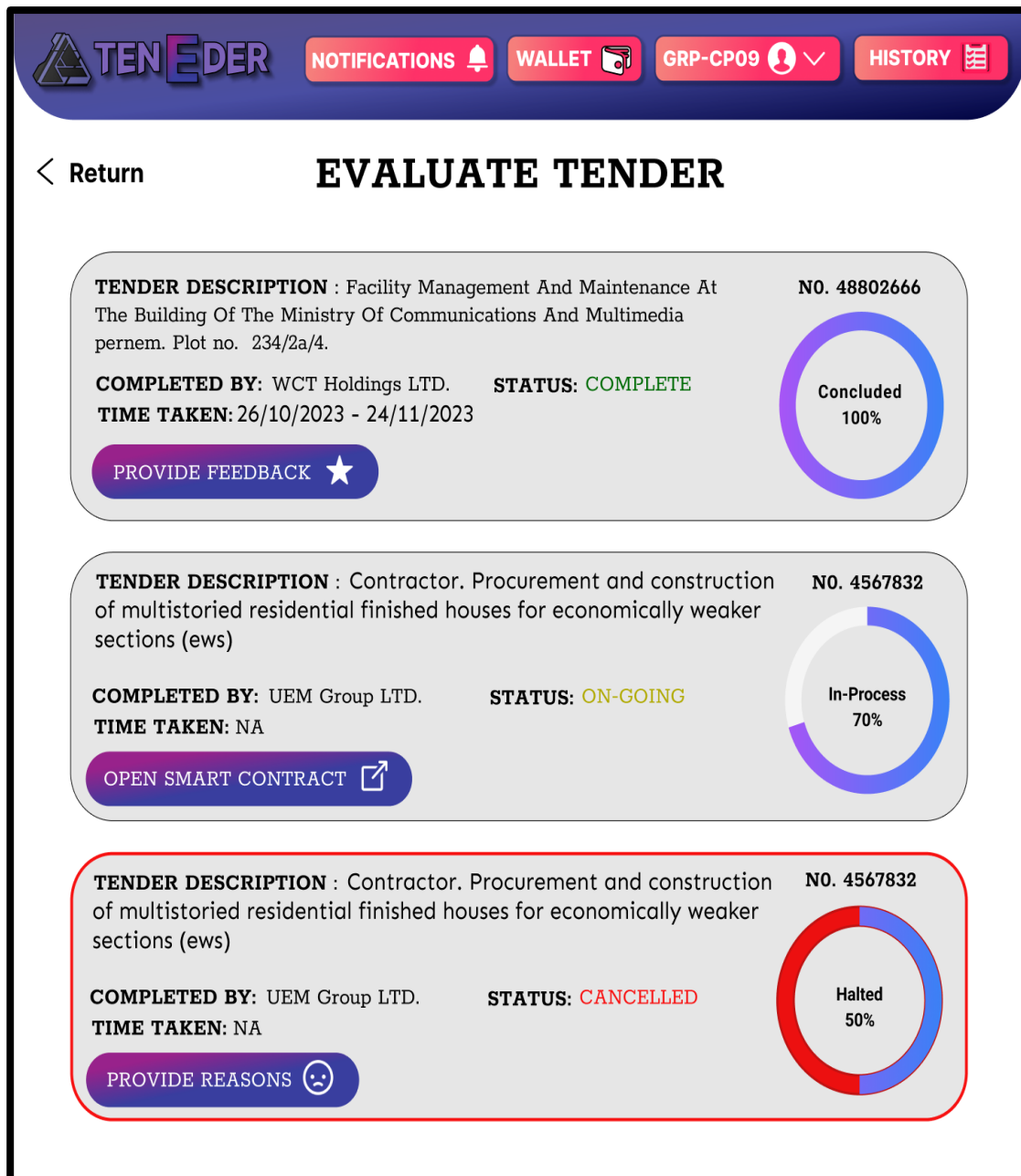


Fig. 5.8 Tender Evaluation for Organization.

The evaluate tender function gives an idea to the organization about the percentage of work completed as shown in Fig. 5.8 , also if the work is completed the organization can provide feedback for the same.

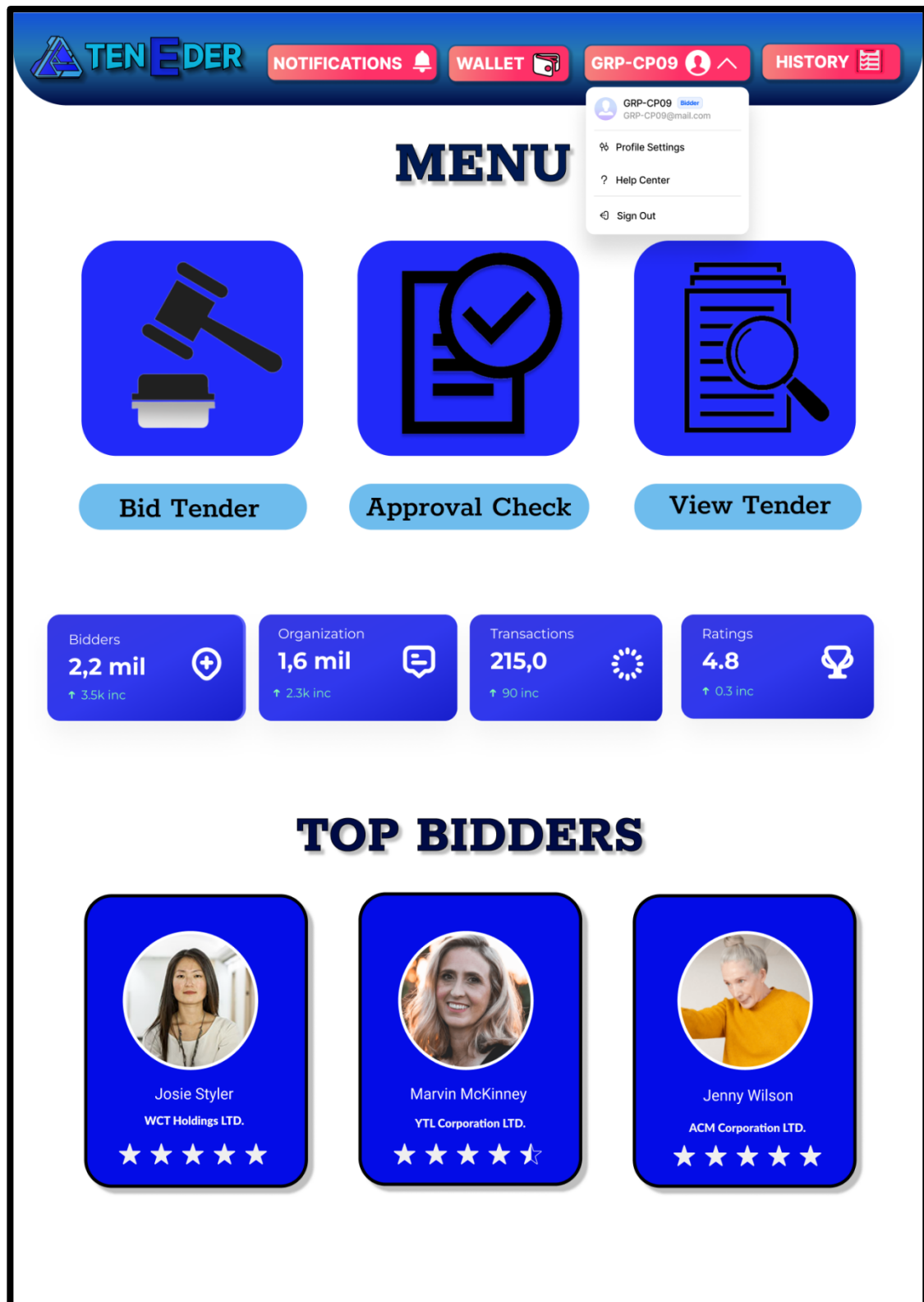


Fig. 5.9 Dashboard for Bidder.

The Fig. 5.12 displays a dashboard containing the various functions that can be performed by the bidder for tender activities. These include placing a bid for a tender, checking for approvals and to view work done related to tender.



NOTIFICATIONS 
WALLET 
GRP-CP09 
HISTORY 

BID TENDER

Keyword search ▼

Type keyword eg-VALUE, DUE DATE etc. 

TENDER DESCRIPTION	TYPE	DUE DATE	BID
1. Contractor. Procurement and construction of multistoried residential finished houses for economically weaker sections (ews). No. 4567832  Panjim  India	Government	22 Days to go 15-Feb-2024	
2. Management Services Of An Apartment House , opp district court. No. 7899020  Mapusa  India	Private	42 Days to go 07-Mar-2024	
3. Facility Management And Maintenance At The Building Of The Ministry Of Communications And Multimedia pernem. Plot no. 234/2a/4. NO. 48802666  Pernem  India	Government	Concluded 	
4. Electrical System Installation For Unimap-construction Of University Of Vasco ,opp Health Center Building (tender No . Pers/r-pk (4) 1511/15-2022) No. 48489530  Vasco  India	Government	88 Days to go 26-Apr-2024	

Fig. 5. 10 Bid Tender for Bidder.

The bid tender fuction gives an opportunity to bidder to view all the tender uploaded by the Organization as shown in Fig. 5.10


NOTIFICATIONS
WALLET
GRP-CP09
HISTORY

< Return

BID DETAILS

Tender Details

Organization : YTL Corporation LTD.

T. Number : 4567832

Description : Contractor. Procurement and construction of multistoried residential finished houses for economically weaker sections (ews).

Tender Motive : Bridge tender aims to unite cities, boost commerce, enhance transportation, and foster regional growth through connectivity.

Key Values

Bidding Period : 21/03/2022 - 21/04/2022

Work Period : 31/04/2022 - 31/07/2022

Location : Panjim

Type : Private

Attachments :  **Municipality_Tender.pdf**

Tender Documents

NIT : Download Upload

Fig. 5.11 Bid Details for Bidder.

The Fig. 5.11 gives the bidder all the details about a particular tender that he wishes to bid for and he can do so by uploading a bid in the form of a document.

NOTIFICATIONS

WALLET

GRP-CP09

HISTORY

APPROVAL CHECK

Keyword search ▼

Type keyword eg-VALUE, DUE DATE etc.

Q

TENDER DESCRIPTION	TYPE	Organization Name	STATUS
1. Contractor. Procurement and construction of multistoried residential finished houses for economically weaker sections (ews). No. 4567832 📍 Panjim 🇮🇳 India	Government	Panjim Municipal Council	SELECTED
2. Management Services Of An Apartment House , opp district court. No. 7899020 📍 Mapusa 🇮🇳 India	Private	Quadros Motors	SELECTED
3. Facility Management And Maintenance At The Building Of The Ministry Of Communications And Multimedia pernem. Plot no. 234/2a/4. NO. 48802666 📍 Vasco 🇮🇳 India	Government	Mormugao Municipal Council	REJECTED
4. Electrical System Installation For Unimap-construction Of University Of Vasco ,opp Health Center Building (tender No . Pers/r-pk (4) 1511/15-2022) No. 48489530 📍 Madgaon 🇮🇳 India	Government	Madgaon Municipal Council	REJECTED

Fig. 5.12 Approval Check for Bidder.

The Bidder will be able to check whether he has been selected or rejected by the Organization for all tenders he had applied for as shown in Fig. 5.12

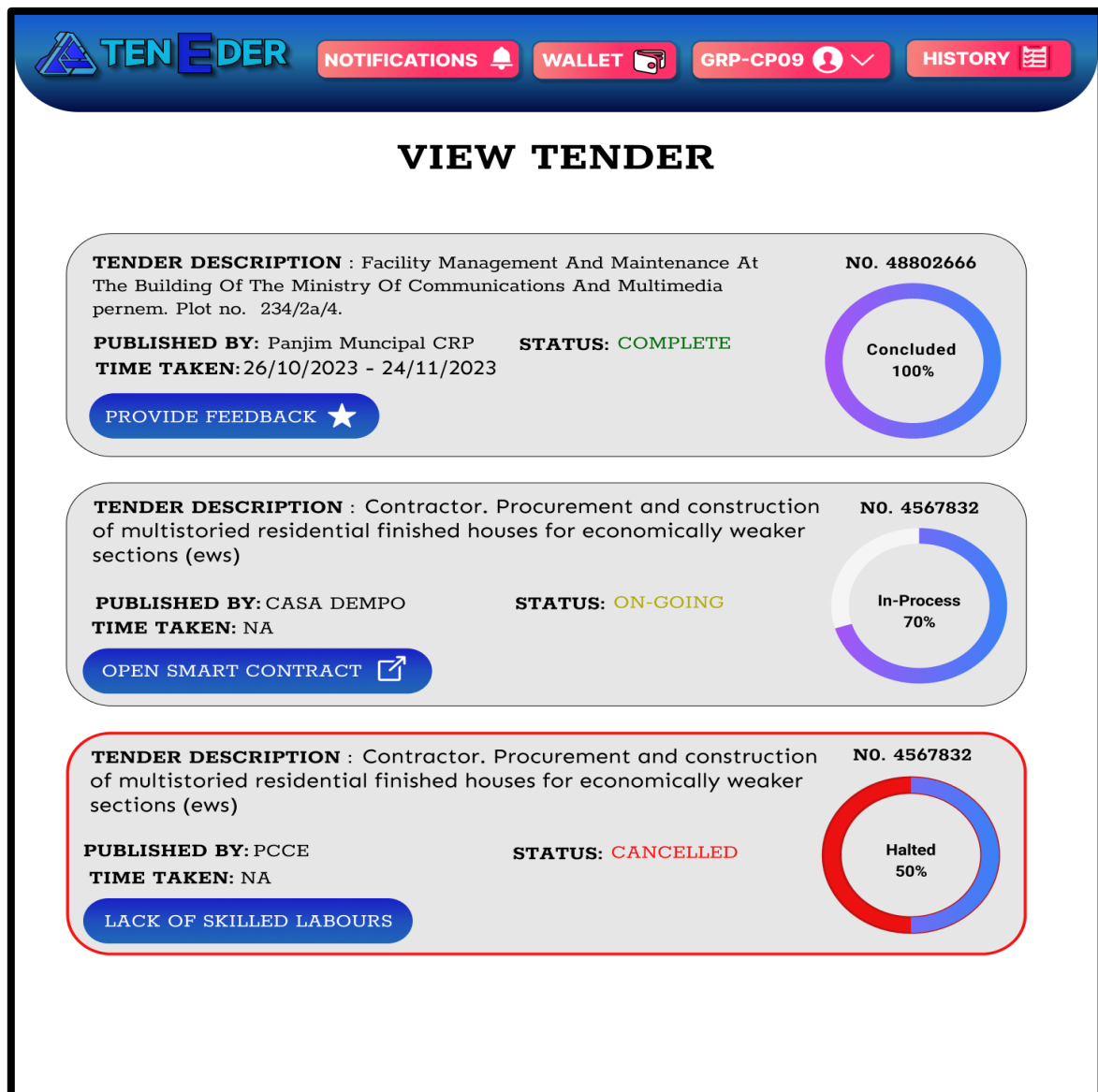


Fig. 5.13 View Tender for Bidder.

The view tender function allows the Bidder to keep a track of the amount of work that has been completed. He can also check if any tender has been cancelled along with the appropriate reason for cancellation as shown in Fig. 5.13

5.2 CHAPTER SUMMARY

In summary, we looked at most of the UI designs. This was created by keeping in mind all the requirements of a basic tender management system. With the help of this, we have implemented the front-end of our project.

CHAPTER 6

RESULTS

In this chapter, we discuss our journey in e-tender management which has been quite enlightening. By using the power of Blockchain and Smart Contracts, we've tackled some significant challenges in how tender processes unfold. One big win has been in transparency. We've managed to give everyone involved a clear view of what's happening in the tendering process. No more shady deals or hidden information. Security-wise, we've stepped up our game too. With blockchain technology, we've made sure that no one can tamper with the data. That means fewer worries about fraud or data breaches. Tasks that used to take ages are now done in a snap. Thanks to Smart Contracts, everything runs smoothly, with minimal room for errors also we discuss the gas requirement for each function that is essential in implementation of smart-contract. As we move forward, we look at the security analysis of the project as well as the proposed system.

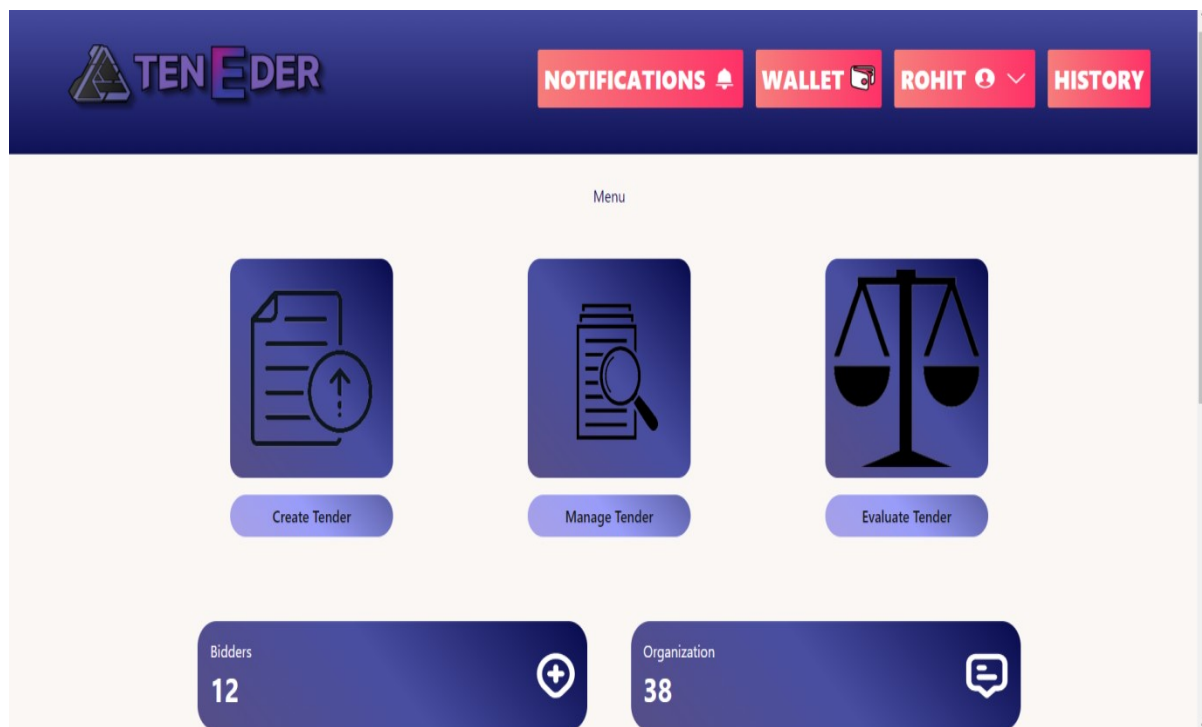


Fig 6.1 Organisation Home Page

The fig 6.1 basically shows all the 3 essential functions of the Organization section which are namely (i) Create tender, (ii) Manage tender and (iii) Evaluate tender. Along with this, the website also allows the organization to view its funds through the Wallet button, also check notifications and to view history of all the previous tenders. The website also shows the current number of bidders and the organizations using the website and also the ratings provided.

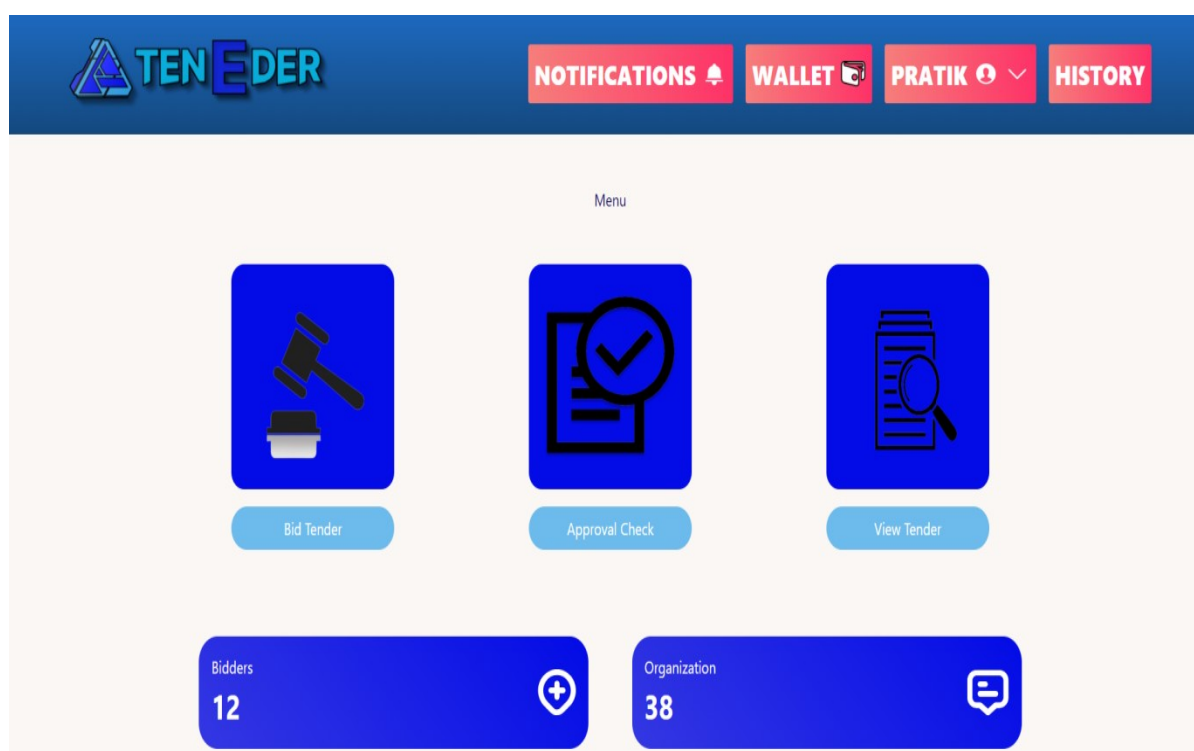
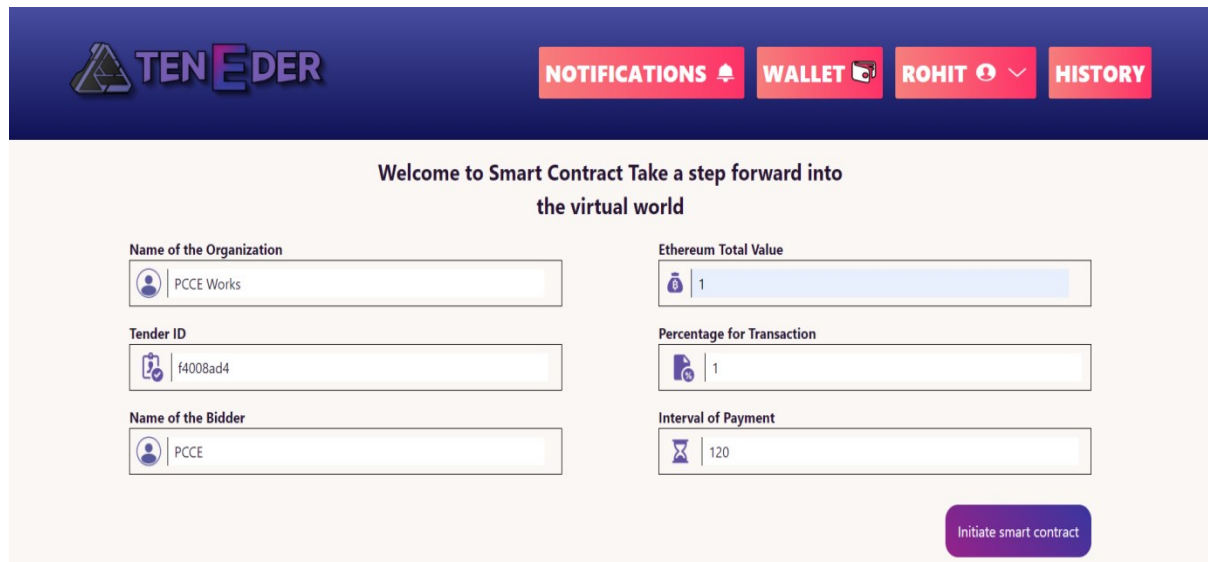


Fig 6.2 Bidder Home Page

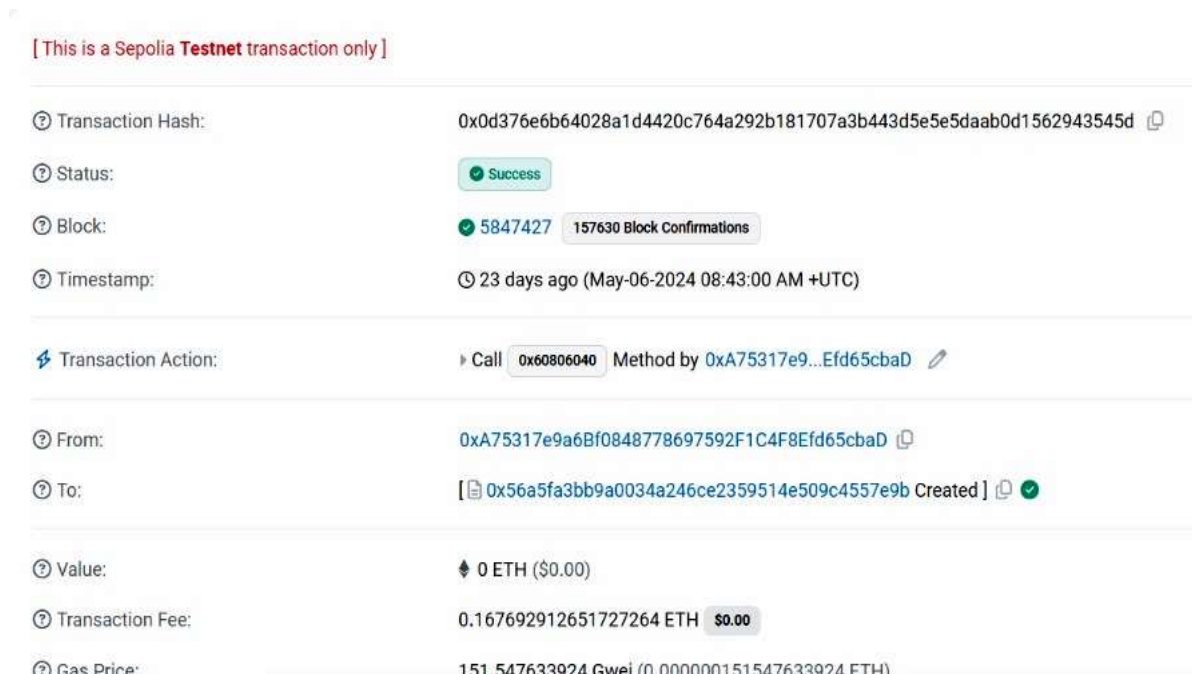
The fig 6.2 basically shows all the 3 essential functions of the Bidder section which are namely (i) Bid tender, (ii) Approval check and (iii) View tender. Along with this, the website also allows the bidder to view its funds through the Wallet button, also check notifications and to view history of all the previous tenders. The website also shows the current number of bidders and the organizations using the website and also the ratings provided.



The screenshot shows the 'TENDER' system interface. At the top, there's a navigation bar with 'NOTIFICATIONS', 'WALLET', 'ROHIT', and 'HISTORY' buttons. Below this, a welcome message reads: 'Welcome to Smart Contract Take a step forward into the virtual world'. The main form is divided into two columns. The left column contains three input fields: 'Name of the Organization' (with a user icon and 'PCCE Works'), 'Tender ID' (with a document icon and 'f4008ad4'), and 'Name of the Bidder' (with a user icon and 'PCCE'). The right column contains three input fields: 'Ethereum Total Value' (with a coin icon and '1'), 'Percentage for Transaction' (with a document icon and '1'), and 'Interval of Payment' (with a clock icon and '120'). A purple button labeled 'Initiate smart contract' is positioned at the bottom right of the form.

Fig 6.3 Smart Contract Deployment

The fig 6.3 shows the smart-contract interaction between the organization and the bidder. In this the organization initiates the smart-contract or in other words it deploys the smart-contract between itself and the bidder. It also contains the 3 essentials factors that are (i) Ethereum total value, (ii) Percentage for transaction and (iii) Interval of payment.



The screenshot displays a transaction confirmation page. At the top, a red warning message states: '[This is a Sepolia Testnet transaction only]'. Below this, the transaction details are listed in a table-like format. The 'Transaction Hash' is '0x0d376e6b64028a1d4420c764a292b181707a3b443d5e5e5daab0d1562943545d'. The 'Status' is 'Success'. The 'Block' is '5847427' with '157630 Block Confirmations'. The 'Timestamp' is '23 days ago (May-06-2024 08:43:00 AM +UTC)'. The 'Transaction Action' is 'Call 0x60806040 Method by 0xA75317e9...Efd65cbaD'. The 'From' address is '0xA75317e9a6Bf0848778697592F1C4F8Efd65cbaD'. The 'To' address is '0x56a5fa3bb9a0034a246ce2359514e509c4557e9b' with a note 'Created'. The 'Value' is '0 ETH (\$0.00)'. The 'Transaction Fee' is '0.167692912651727264 ETH \$0.00'. The 'Gas Price' is '151.547633924 Gwei (0.000000151547633924 ETH)'. Each row has a circular icon with a question mark on the left and a link icon on the right.

Fig 6.4 Successful transaction

The fig 6.4 portrays a successful transaction in creating a contract between the organization and the bidder.

SMART-CONTRACT-BASED TENDER MANAGEMENT SYSTEM USING BLOCK-CHAIN

Seppolia Testnet

Search by Address / Txn Hash / Block / Token

Latest 9 from a total of 9 transactions

Download Page Data

Transaction Hash	Method	Block	Age	From	To	Value	Txn Fee
0x5179499456...	Transfer	5995272	37 hrs ago	0x7Ed74647...0F7FB271f	IN 0xA75317e9...Efd65cbaD	0.1 ETH	0.0007701
0x47395fa2a9...	Transfer	5988387	2 days ago	0x631E9B03...68a668d20	IN 0xA75317e9...Efd65cbaD	0.1 ETH	0.00526518
0x3768609c22...	Transfer	5956062	7 days ago	0xA7E4EF0a...974ECD60b	IN 0xA75317e9...Efd65cbaD	0.1 ETH	0.00077655
0xe302608233...	Transfer	5936890	10 days ago	0xA7E4EF0a...974ECD60b	IN 0xA75317e9...Efd65cbaD	0.5 ETH	0.00149822
0x503d270c77...	Transfer	5920473	12 days ago	0xA6023AfB...F41E9d05B	IN 0xA75317e9...Efd65cbaD	0.5 ETH	0.00710835
0xab607ba66c...	Transfer	5893249	16 days ago	0x3C352eA3...6eCC21917	IN 0xA75317e9...Efd65cbaD	0.5 ETH	0.01469172
0xb5ba6d1dc8...	Transfer	5893248	16 days ago	0x87c9B02A...9CF3cd9Bf	IN 0xA75317e9...Efd65cbaD	0.5 ETH	0.01469172
0x0d376e6b64...	0x60806040	5847427	23 days ago	0xA75317e9...Efd65cbaD	OUT Contract Creation	0 ETH	0.16769291
0xa17de408cb...	Transfer	5846524	24 days ago	0x87c9B02A...9CF3cd9Bf	IN 0xA75317e9...Efd65cbaD	0.5 ETH	0.0008085

Fig 6.5 Permanent ledger

The fig 6.5 shows the permanent trail of records of the entire transaction right from creation of contract to distribution of funds.

6.1 BLOCKCHAIN

A blockchain is a decentralized, distributed ledger technology that securely records and verifies transactions across a network of computers. Transactions are grouped into "blocks" and linked together in chronological order, forming a continuous chain. Each block contains a cryptographic hash of the previous block, ensuring the integrity of the data and making it nearly impossible to alter past transactions without altering subsequent blocks. This tamper-resistant nature eliminates the need for intermediaries like banks or governments to verify transactions, reducing costs and enhancing trust. Blockchain technology is transparent, as every participant in the network has access to the same information, yet secure, as transactions are encrypted and stored across multiple nodes. It finds applications in finance, supply chain management, healthcare, voting systems, and more, offering improved efficiency, transparency, and security compared to traditional centralized systems.

6.2 GAS REQUIREMENTS

In blockchain networks, natural gas (often referred to as "gas") plays an important role in supporting transactions and executing smart contracts. It is used as a unit of measurement to represent the computer hardware required to perform these tasks. Transactions and smart contracts on platforms like Ethereum are made using Gas, which is paid with Ethereum users' cryptocurrency. Fuel prices vary depending on the complexity and combination of the operation. Users can create gas limits to determine the maximum amount of gas they want to use for work, thus protecting themselves from unexpected costs. Additionally, users can create gas prices by deciding how much they are willing to pay for a unit of gas to encourage miners to push the business forward. Test networks play an important role in fuel testing. They provide developers with a sandbox environment to deploy and test smart contracts and transactions without using real cryptocurrencies. By using the testnet, developers can simulate

real-life scenarios and measure the fuel needs of their applications without real costs. This allows them to increase fuel efficiency and improve economy, especially during network operations.

Some aspects in which gas will be required in our project is as follows:

- If an organization wants to create a smart-contract between themselves and the bidder.
- If an organization wants to deposit funds into the smart contract.
- If an organization wants to distribute a percentage of funds to the bidder.

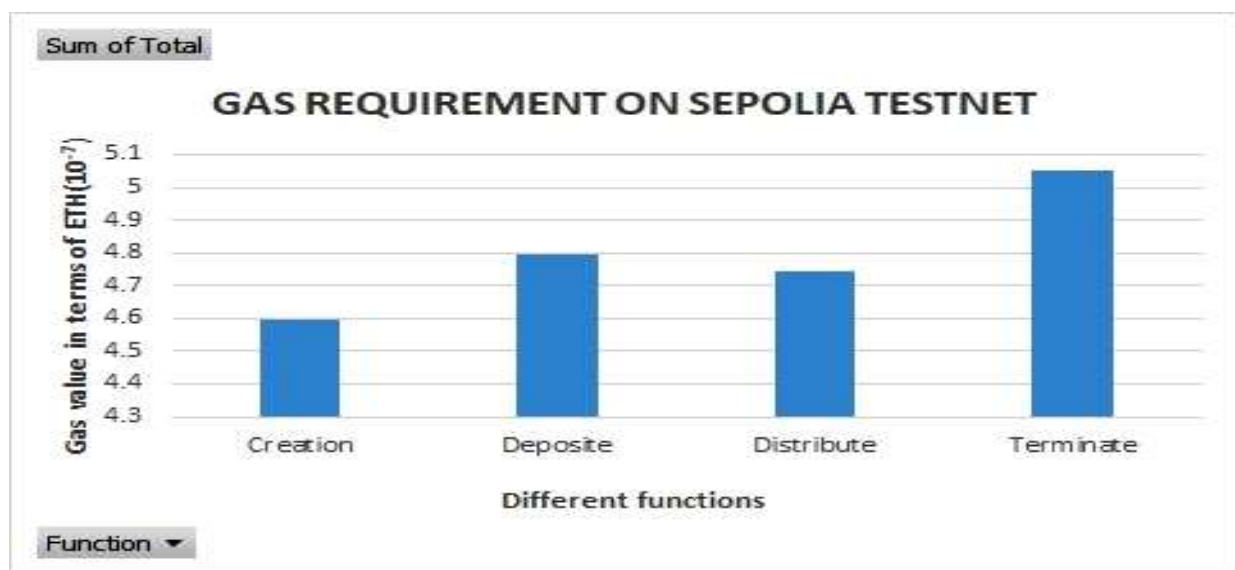


Fig 6.2.i Gas Requirements

Gas Requirement refers to the complexity of the function and is directly proportional to the complexity of the transaction to the blockchain. The Fig 6.2.i shows the Gas Requirements of various functions of the system. In this the distribute function requires the highest amount of gas whereas the initiate function requires the least amount of gas.

6.3 SECURITY ANALYSIS

The security and transparency provided by smart-contract and blockchain are as follows:

- The bidders that place their bids are not viewable to other bidders and even the organization until the due date has lapsed, thus, ensuring a free and fair judgement to all.

- The blockchain network is secure such that once a bidder places its bid it cannot change it until the tender results are out.
- The organizations cannot change the tender requirements (details such as tender_motive, tender_materials, date_of_bidding) so as to favour a particular bidder.
- Smart-contracts ensure that there is a valid contract in place and helps in automating the entire process by basically taking inputs such as percentage of funds to be transferred and when to be transferred, thereby, making sure the organization and bidders transactions go smoothly.

6.4 SMART CONTRACTS

A smart-contract is a computer program that automatically executes and enforces the terms of an agreement or contract when predefined conditions are met. Built on blockchain technology, smart contracts enable trustless and transparent transactions without the need for intermediaries. They operate based on a predetermined set of rules and conditions written into code, ensuring parties' compliance without relying on third-party enforcement. Smart contracts can facilitate a wide range of transactions, including financial agreements, supply chain management, digital identity verification, and more. Their decentralized nature enhances security and reduces the risk of fraud or manipulation. Once deployed on the blockchain, smart contracts are immutable and tamper-resistant, providing a reliable and efficient way to conduct business in various sectors.

→ **FUNCTION TO DEPOSIT FUNDS INTO THE SMART-CONTRACT**

```
function depositFunds() public payable onlyOwner {  
    require(msg.value == totalAmount, "Incorrect amount sent");  
}
```

→ **FUNCTION TO DISTRIBUTE FUNDS TO THE BIDDER**

```
function distributeFundsAutomatically() public onlyOwner {
    require(block.timestamp >= startTime, "Payment period has not started yet");
    require(totalPaid < totalAmount, "All funds have been distributed"); }
```

→ **FUNCTION TO CHECK IF THE CORRECT AMOUNT HAS BEEN DEPOSITED**

```
function checkDeposit() public view returns (bool) {
    return address(this).balance == totalAmount;
}
```

→ **FUNCTION TO CALCULATE AND UPDATE WHEN THE NEXT DISTRIBUTION TO OCCUR**

```
uint256 nextDistributionTime = lastDistributionTime + paymentIntervalInSeconds;
}
lastDistributionTime = block.timestamp;
}
function manualTrigger() public onlyOwner {
    distributeFundsAutomatically();
}
```

→ **FUNCTION TO DENY PAYMENT AND REVERT ALL THE FUNDS IN THE ORGANIZATION ACCOUNT**

```
function withdrawExcess() public onlyOwner {
    require(block.timestamp >= startTime, "Payment period has not started yet");
    require(totalPaid >= totalAmount, "Total amount not paid yet");
    owner.transfer(address(this).balance - totalAmount);
}
```

6.5 PROPOSED SYSTEM

Based on the below proposed system (fig 7.4.I), we elaborate the working of the entire project right from the Bidder registration to checking of balance in e-wallet.

System Overview

The proposed system integrates multiple stakeholders, including bidders and organizations, into a unified framework facilitated by a blockchain. The system manages the entire lifecycle

of a tender, from creation to final feedback, ensuring transparency, security, and immutability of records.

Stakeholders

1. Bidder: An entity that participates in the tender process by submitting bids.
2. Organization: The entity that creates and manages tenders.
3. System: The intermediary that facilitates interactions between bidders and organizations.
4. Blockchain: The underlying technology ensuring the security and immutability of tender-related transactions.

Key Processes

1. Registration and Authentication

- Bidder/Organization Registration:

- Both bidders and organizations must register in the system.
- Upon registration, they receive unique credentials for secure access.

- Login:

- Registered users (bidders and organizations) log into the system using their credentials.

- Forgot Password:

- Users can request a password reset if they forget their login credentials.

2. Tender Creation and Notification

- Create Tender (Organization):

- The organization creates a new tender, specifying details such as scope, requirements, and deadlines.
- The system sends notifications to potential bidders about the new tender.

3. Bid Submission and Notification

- Bid Tender (Bidder):
 - Interested bidders submit their bids via the system.
 - The system notifies the organization of the received bids.

4. Tender Management

- Manage Tender (Organization):
 - The organization reviews submitted bids and can:
 - Cancel pre-tender - Select a bid - Reject a bid
 - Notifications are sent to the respective bidders based on the action taken.

5. Approval and Smart Contract Initiation

- Approval Check (System):
 - The system verifies the status of submitted bids:
 - Pending - Cancelled - Rejected - Selected
 - Bidders can check reasons for cancellation or rejection.
- Initiate Smart Contract (Organization):
 - Upon selection of a bid, the organization initiates a smart contract on the blockchain, defining the terms and conditions.

6. Tender Evaluation and Feedback

- Evaluate Tender (Organization):
 - The organization evaluates the progress of the tender.
 - Feedback is provided at various stages:
 - Work in progress - Work completed
 - Reasons are provided if the tender is cancelled post-award.

7. Smart Contract Execution

- Open Smart Contract (Bidder/Organization):
 - Both parties can open and review the smart contract to check the status:

- Work in progress - Work completed
- Feedback is exchanged upon completion of work.

8. Wallet Check

- Check Wallet (Bidder/Organization):

- Both parties can check their respective wallets to verify payments and transactions made via the smart contract.

Security and Transparency

- Blockchain Integration:

- All critical actions and transactions are recorded on the blockchain.
- Ensures tamper-proof records and transparency throughout the tender process.

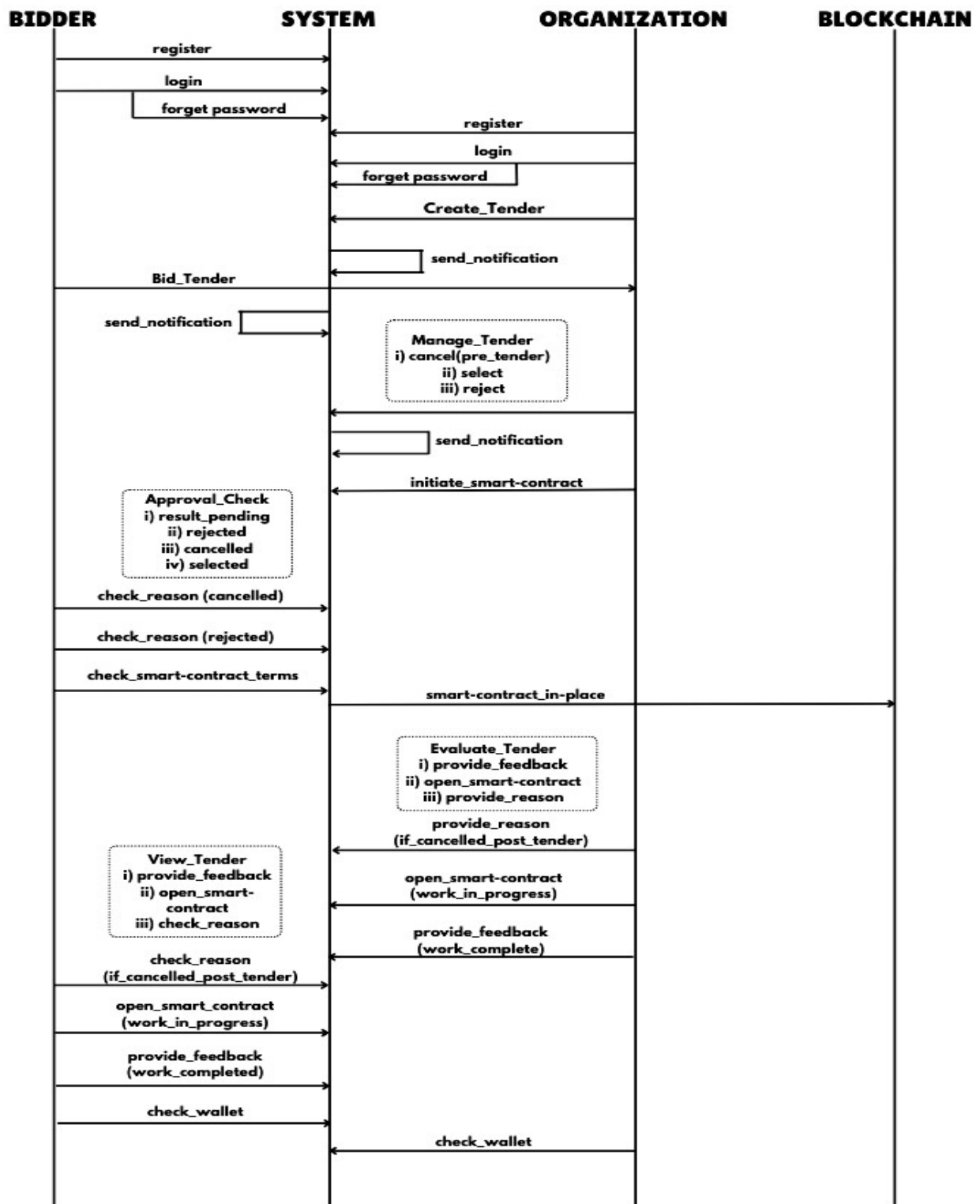


Fig. 6.5.i Proposed System

6.6 CHAPTER SUMMARY

In this chapter we looked at all the implementations and functions that we have done and researched upon to make our project successful. This chapter gives us the basic of how to go about the project and also gives us an idea of some of the hidden or unknown features that run in the background.

CHAPTER 7

FUTURE SCOPE

In this chapter we look at the future advancements that could be made to our project so that this project could be improvised and drawbacks can be prevented. We're keen to improve our system and explore new tech to make tender-process even better. Our aim is to keep improving transparency, security, and efficiency in the world of tendering, making life easier for everyone involved.

The future of the blockchain-based smart contract tender management system involves several key improvements. First, it needs to handle more transactions by using techniques like Layer 2 solutions and sharding. Making the system work well with other blockchain networks and standardizing smart contract protocols will also be important. To keep the system secure, new encryption methods and protections against future threats will be necessary.

Making the system easier to use and more accessible is crucial. This can be achieved by creating a mobile app and offering user training programs. Adding artificial intelligence can help predict outcomes and detect fraud. Ensuring the system meets different legal requirements will also help it be adopted more widely.

Finally, expanding the system's capabilities by connecting it with Internet of Things (IoT) devices and decentralized finance (DeFi) platforms will make the tendering process even better. These improvements will help the blockchain-based tender management system grow and become more effective.

CONCLUSION

Our project "Smart-Contract-Based Tender Management System Using Blockchain" aims to make tender processes simpler and more trustworthy. By using blockchain, the system ensures that every step of the tender process is recorded and can't be changed later, making it fairer for everyone involved.

Throughout this journey, we faced some challenges, like making sure our system followed all the rules and regulations and ensuring that everyone could easily use it. But these challenges only fueled our commitment to making a positive change. As we conclude, it's not just about this project; it's about showing that combining innovative technology, like blockchain, with regular processes can lead to better, more reliable systems. This project is a step towards a future where technology helps us work together more effectively and transparently.

In wrapping up, it's exciting to see how technology can truly transform everyday processes. By using blockchain, we aimed to make the tendering system more open, trustworthy, and faster. The idea was to create a system where everyone involved, from those issuing tenders to those bidding for them, could benefit from a secure and efficient platform.

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